Chronic Inflammation

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A Quick Recap



- Name cells which secrete Histamine and Serotonin
- Which mediators causes pain
- What is serous inflammation in body cavities termed as
- What happens if fibrin is not removed in fibrinous inflammation
- What is localized collection of neutrophils and cellular debris termed as
- Persistence of etiological agent and recurrent infection in acute inflammation leads to?

A Quick Recap



- Mast cells, Basophils, Platelets
- PGE2, Bradykinin
- When occur in body cavities termed as EFFUSSION
- If the fibrin is not removed, over time it may stimulate the growth of fibroblasts and blood vessels and thus lead to scarring (organization)
- Abscess
- Chronic inflammation

Objectives



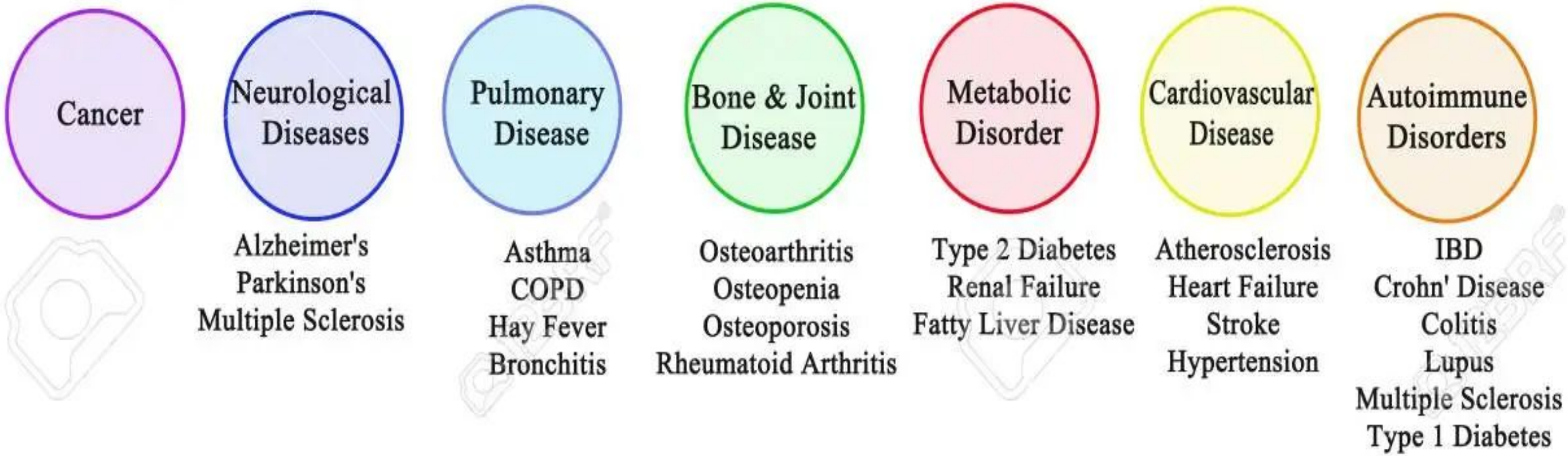
At the end of the lecture students should be able to:

- Define chronic inflammation
- Name causes of chronic inflammation
- Enumerate cells involved in chronic inflammation
- Describe functions of different cells involved in chronic inflammation

Chronic Inflammation

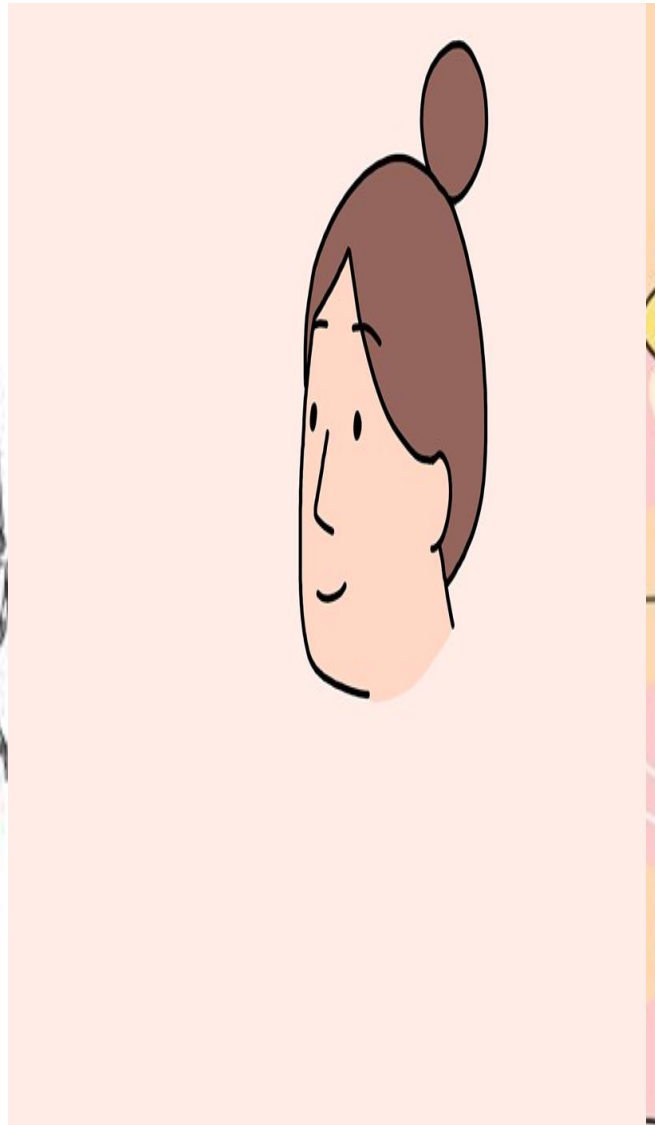
Chronic inflammation is a response of prolonged duration (weeks or months) in which inflammation, tissue injury and attempts at repair coexist, in varying combinations.

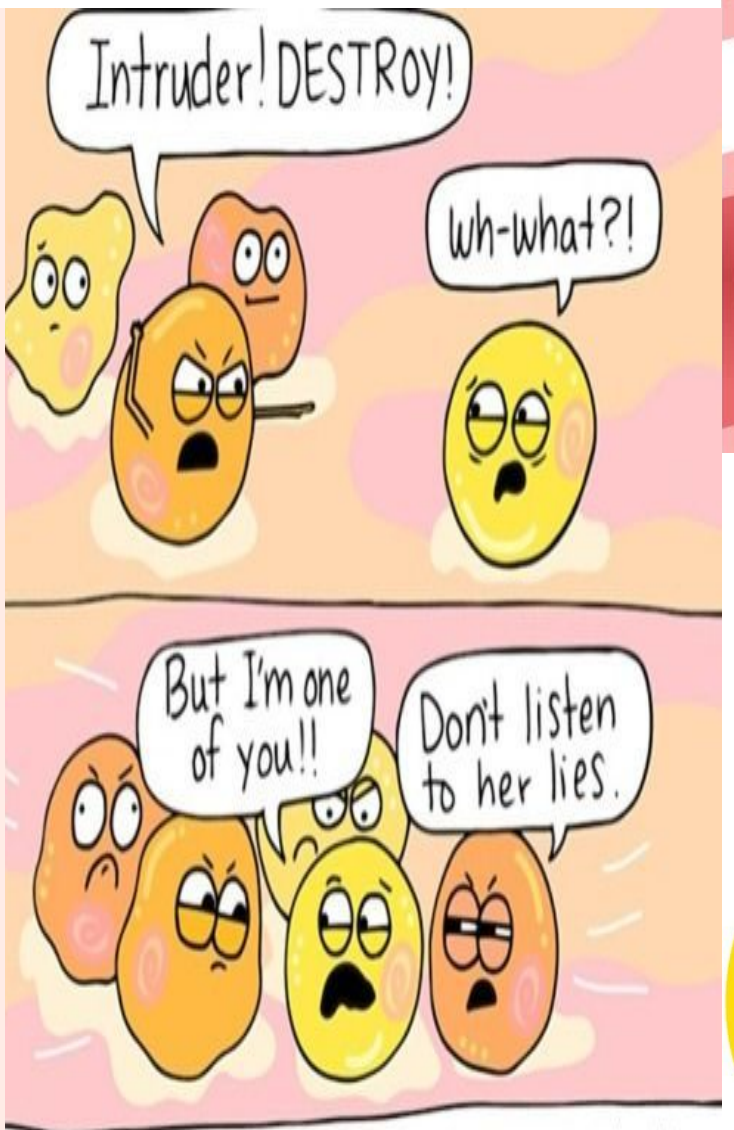
It may follow acute inflammation, may begin insidiously

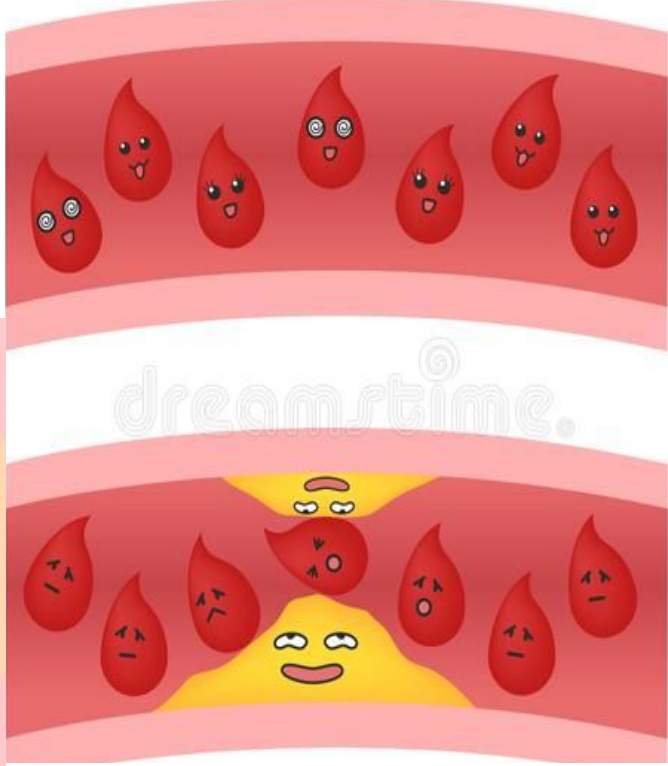


Causes of Chronic Inflammation??



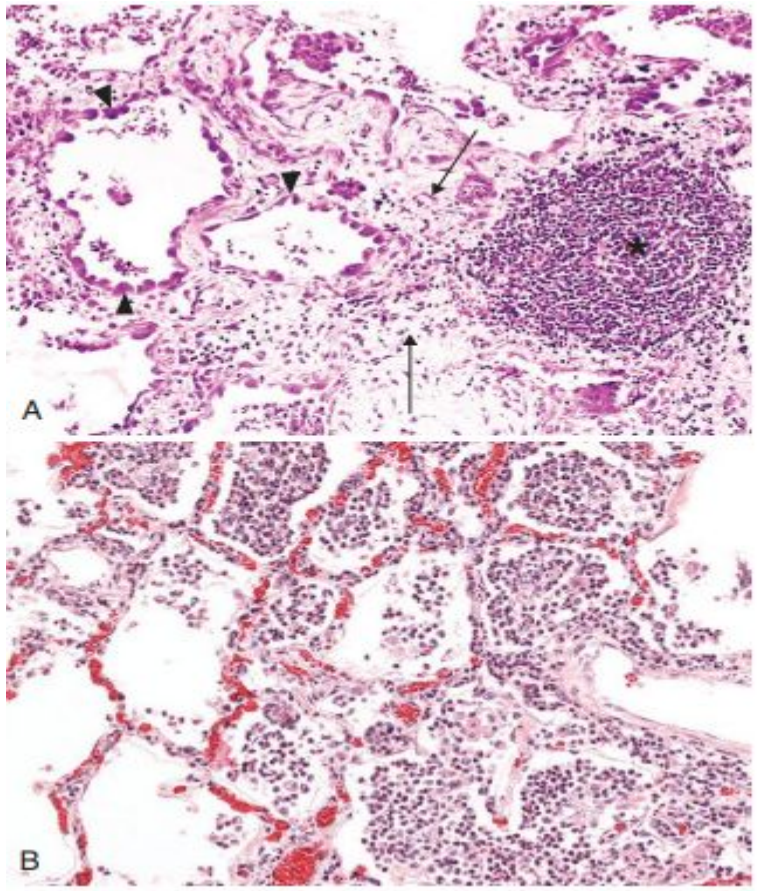




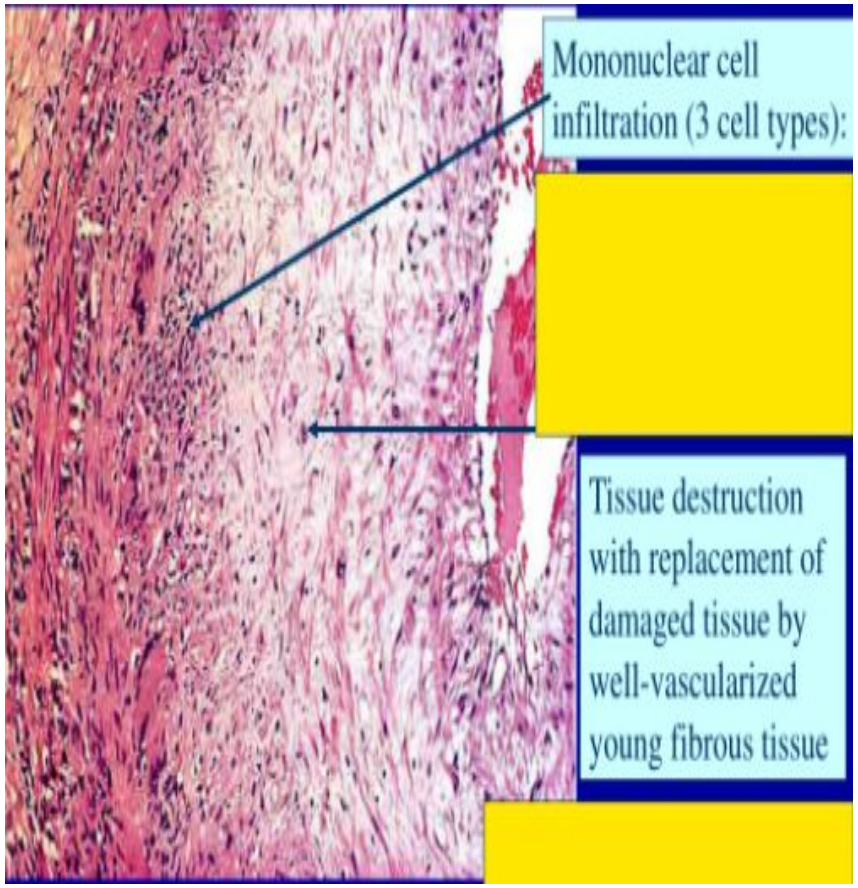




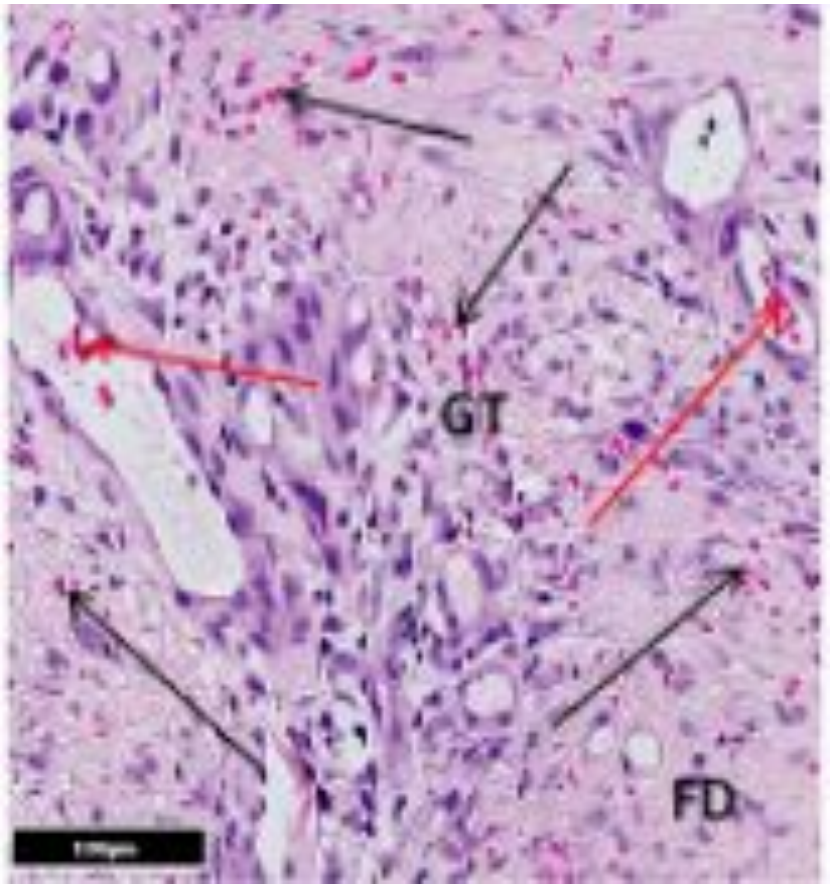
Morphological Features



Infiltration with mononuclear cells

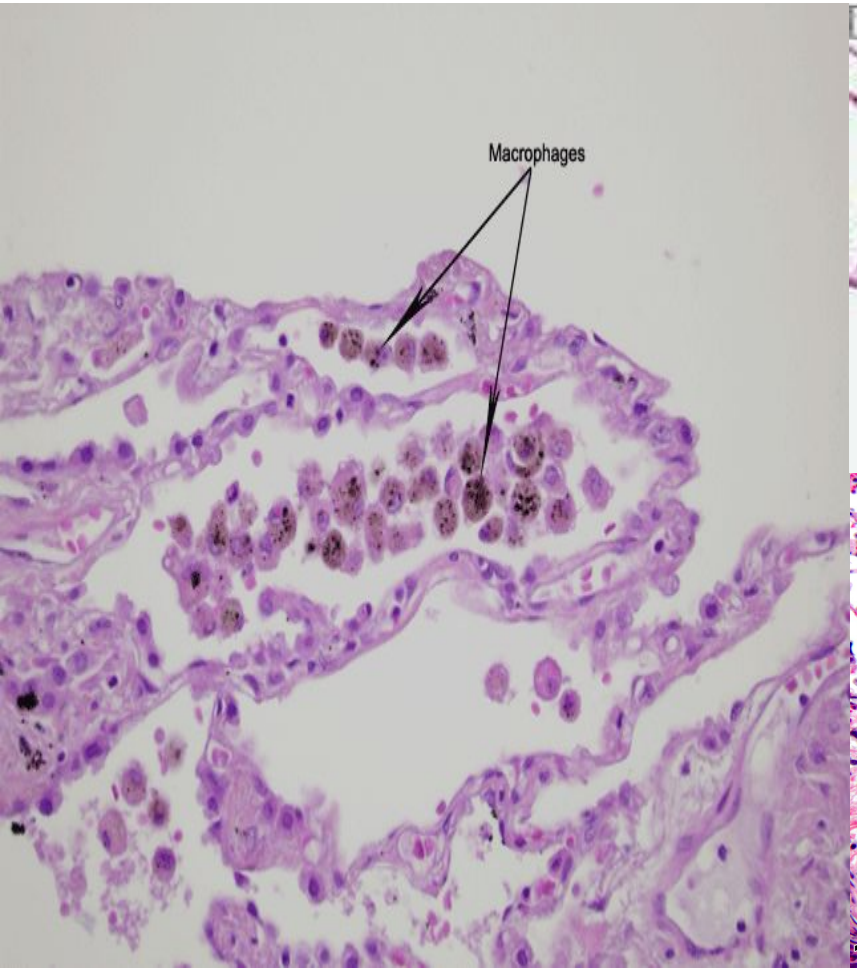


Tissue destruction

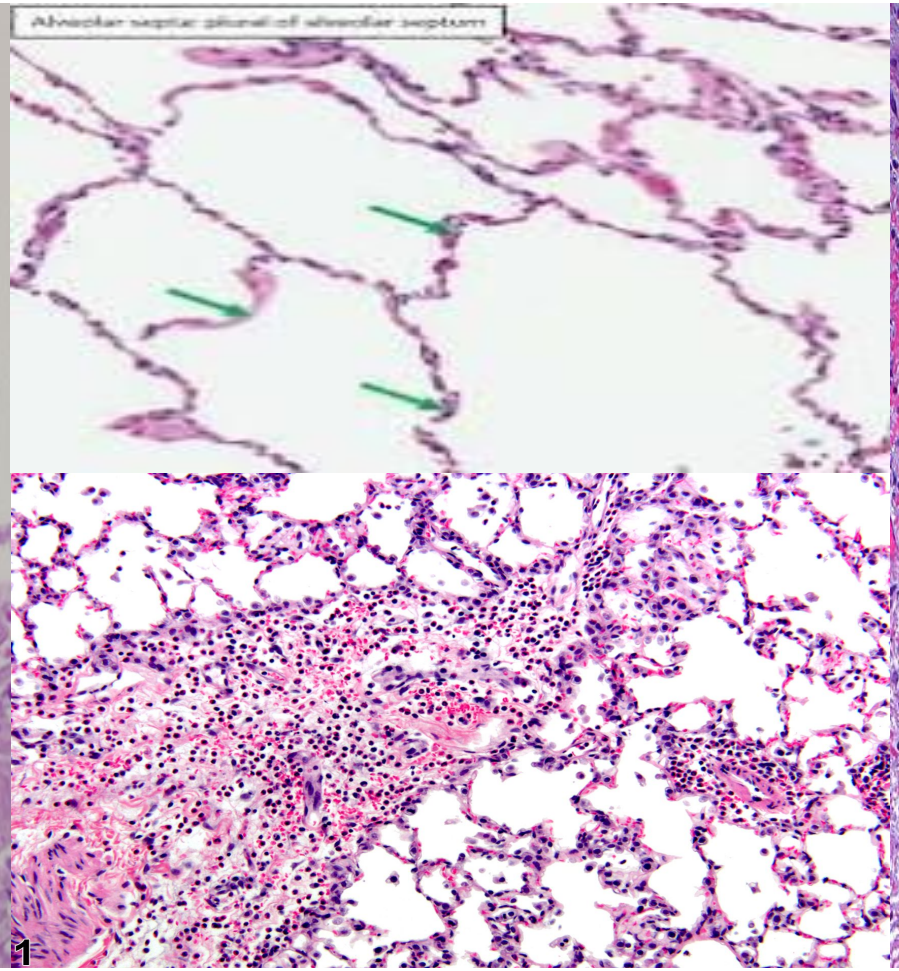


Attempts at healing

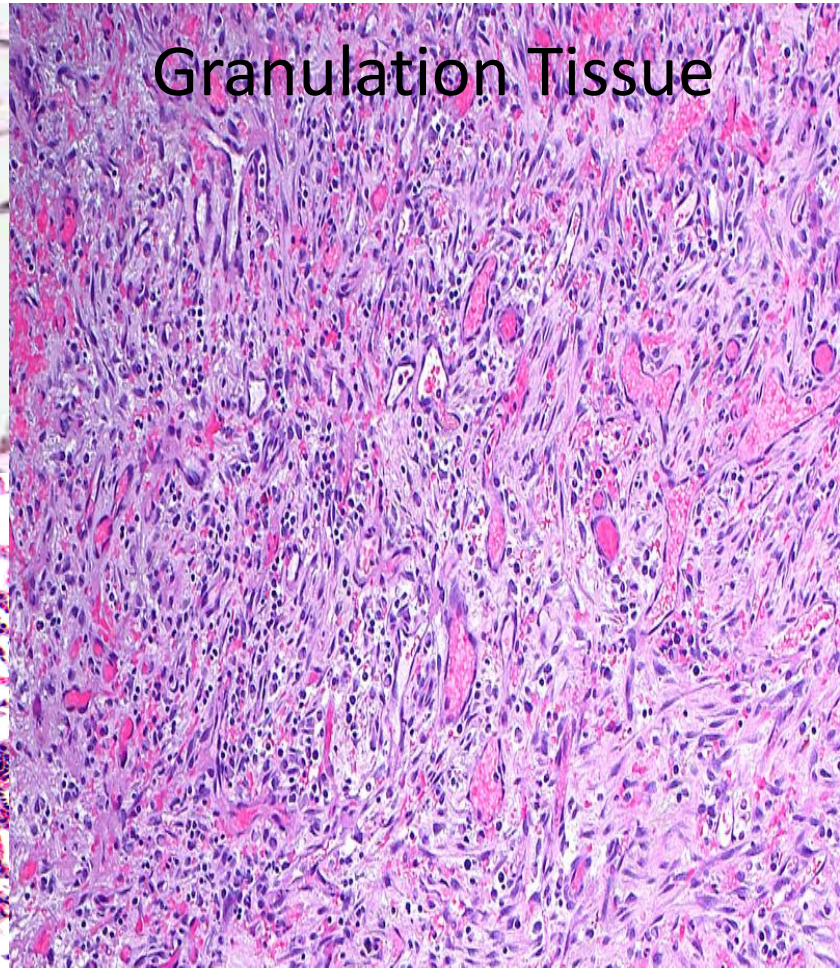
Morphological Features



Infiltration with mononuclear cells



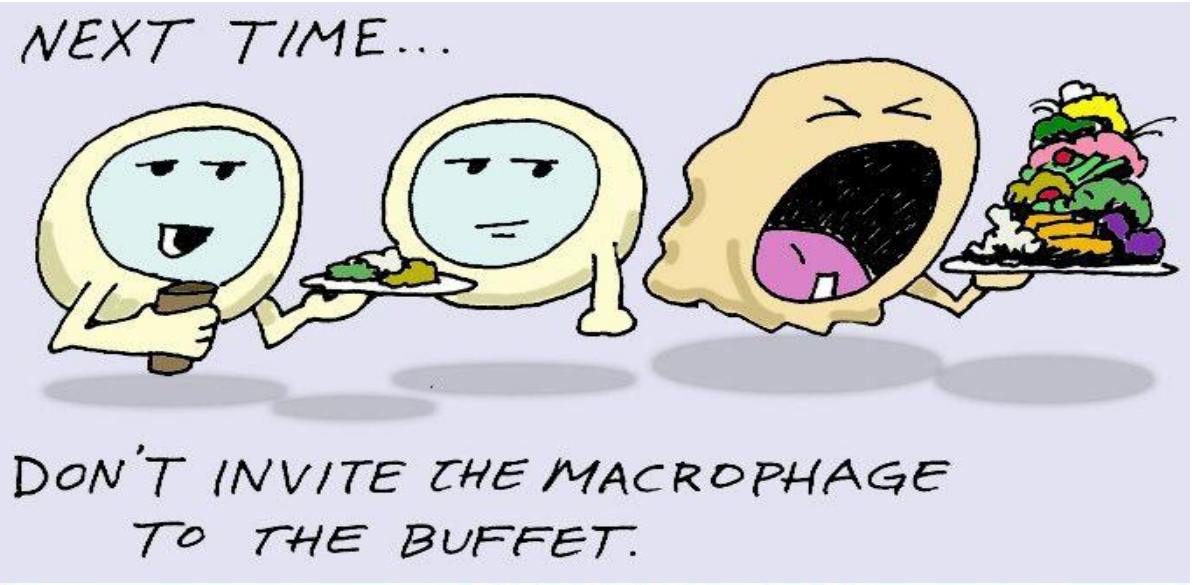
Tissue destruction

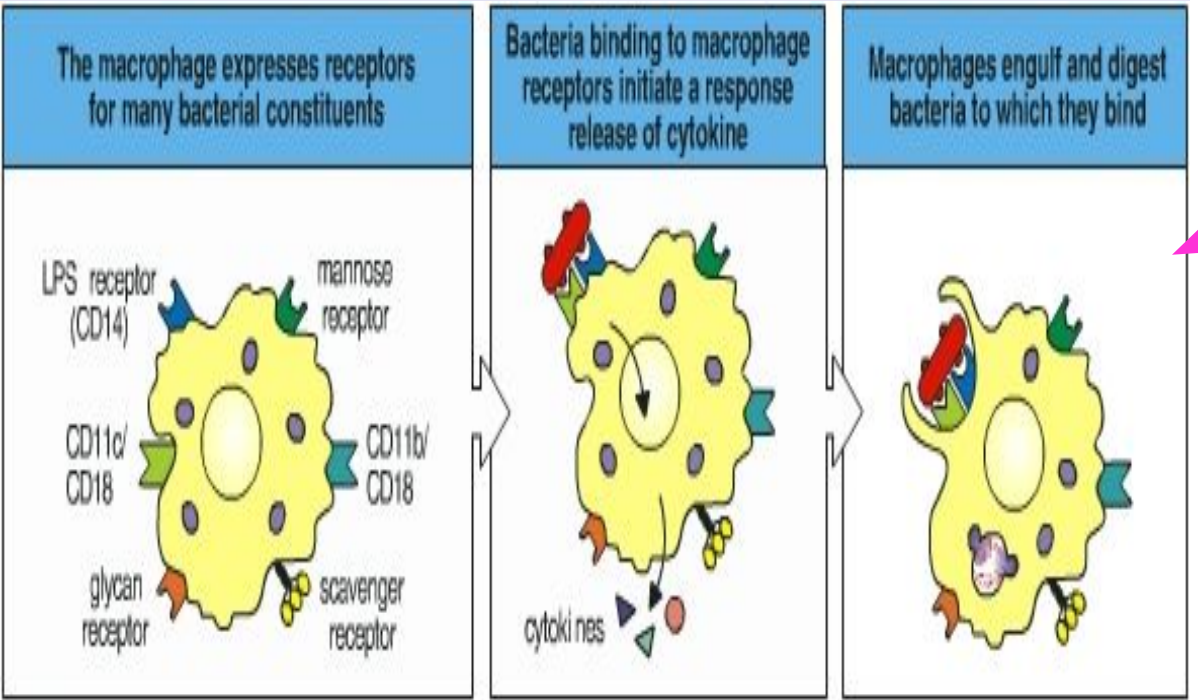


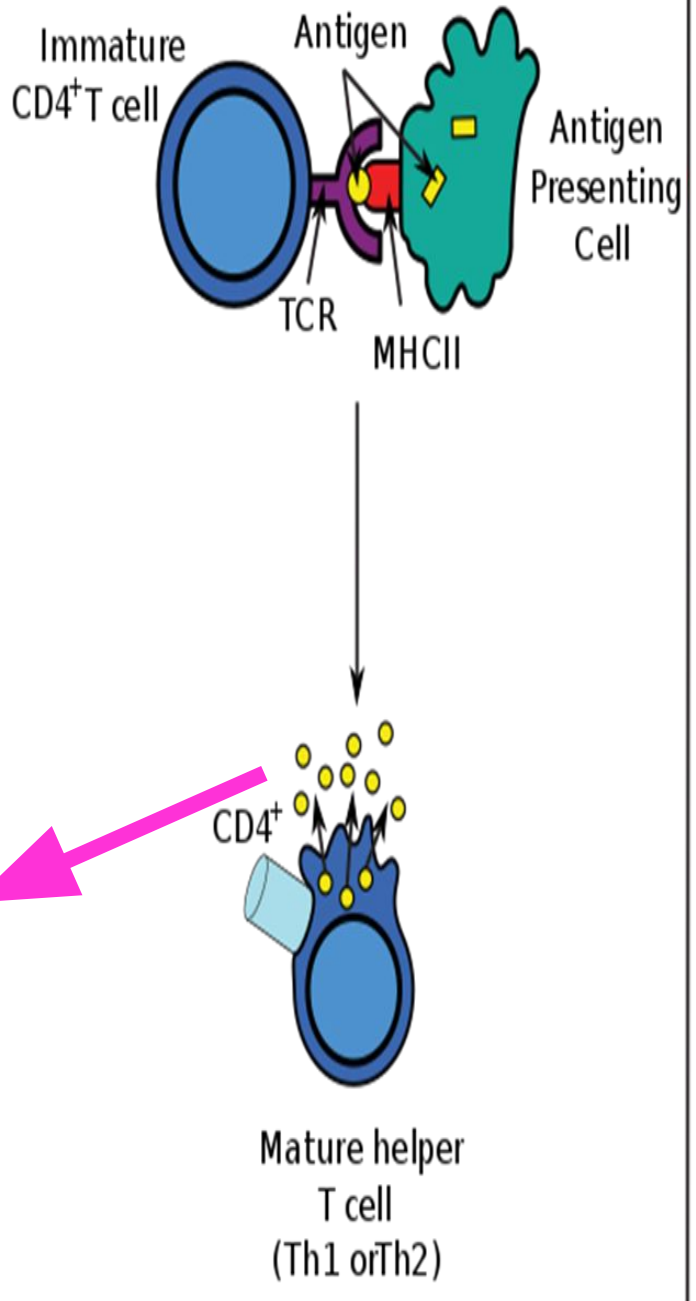
Attempts at healing

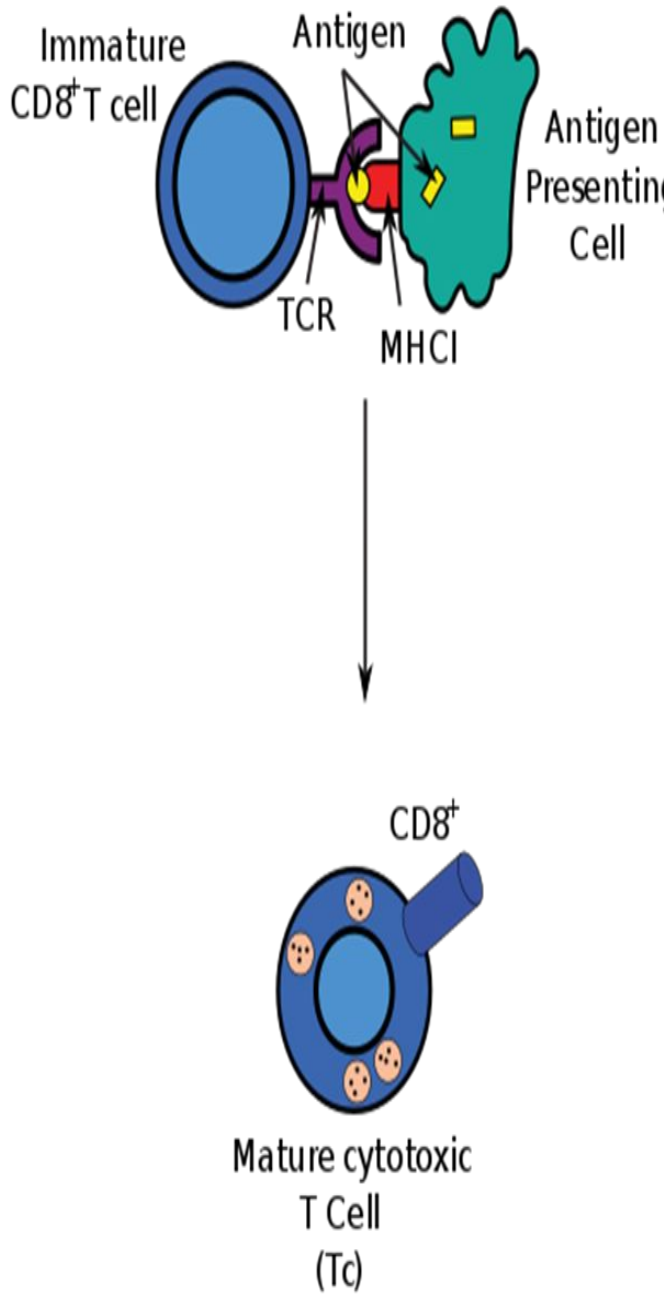
Cells of Chronic Inflammation

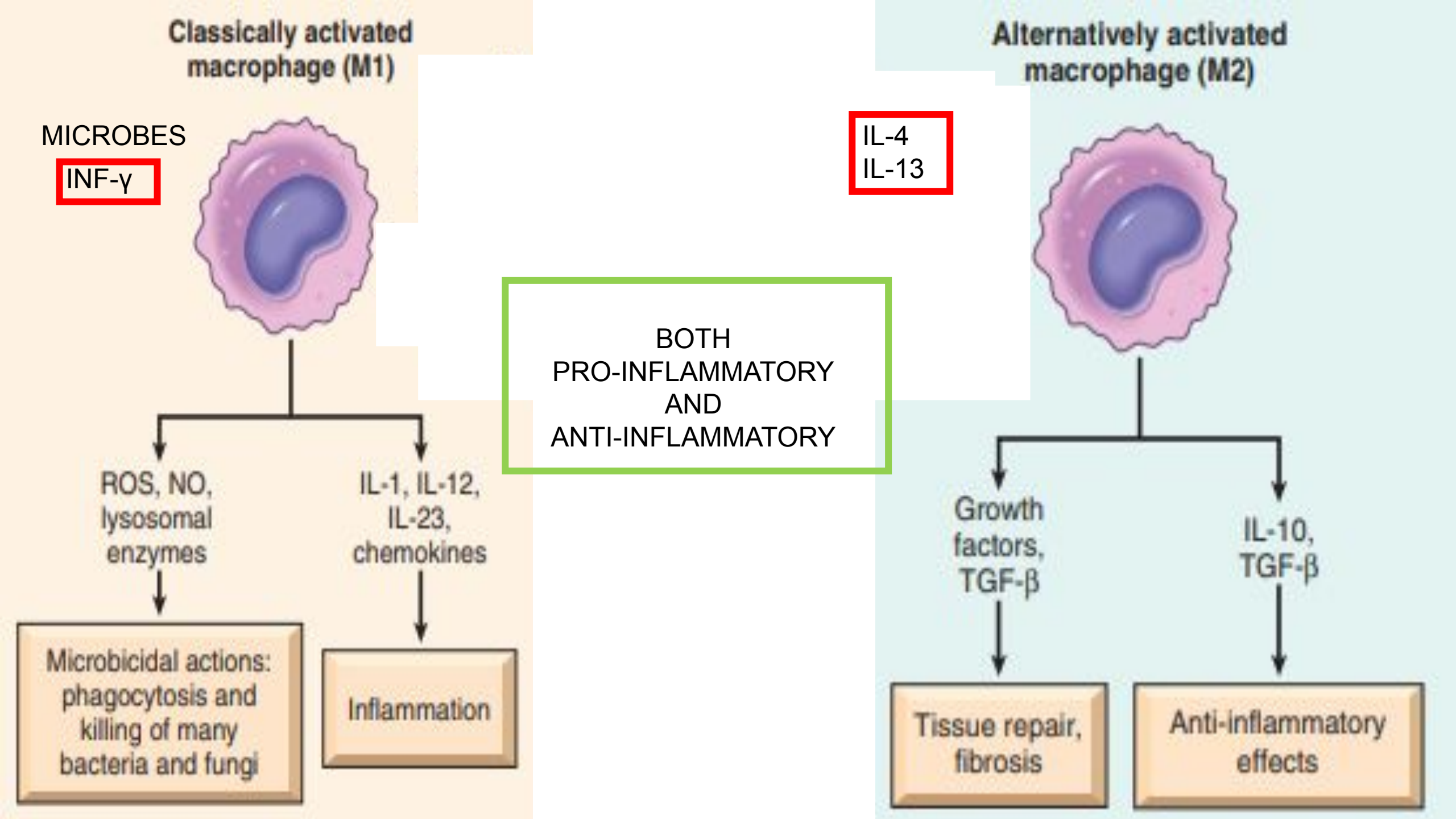
Macrophage


NEXT TIME...
DON'T INVITE THE MACROPHAGE TO THE BUFFET.

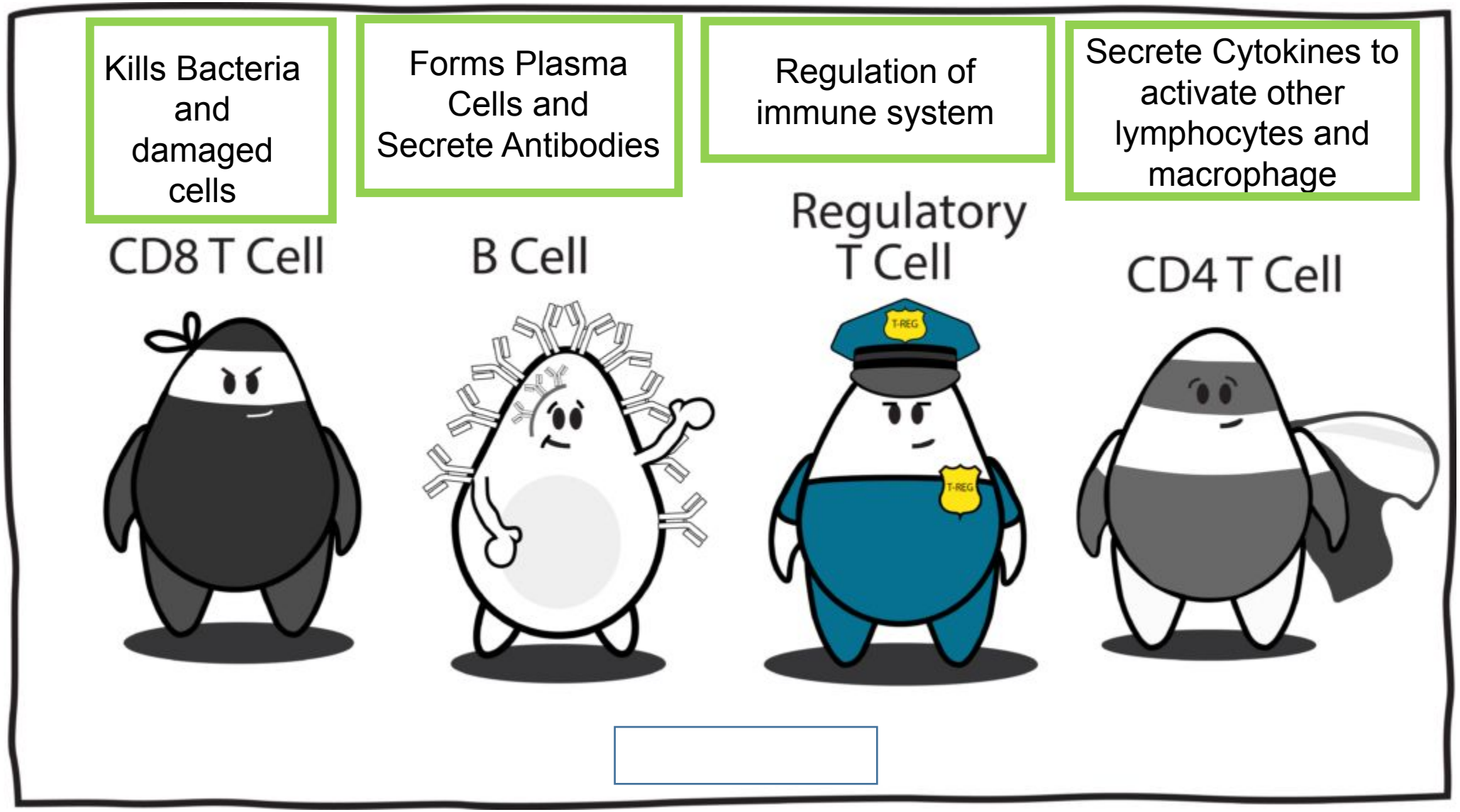

The macrophage expresses receptors for many bacterial constituents
Bacteria binding to macrophage receptors initiate a response release of cytokine
Macrophages engulf and digest bacteria to which they bind
LPS receptor (CD14)
mannose receptor
CD11c/ CD18
CD11b/ CD18
glycan receptor
scavenger receptor
cytokines


Immature CD4+ T cell
Antigen
Antigen Presenting Cell
TCR
MHCII
CD4+
Mature helper T cell (Th1 orTh2)

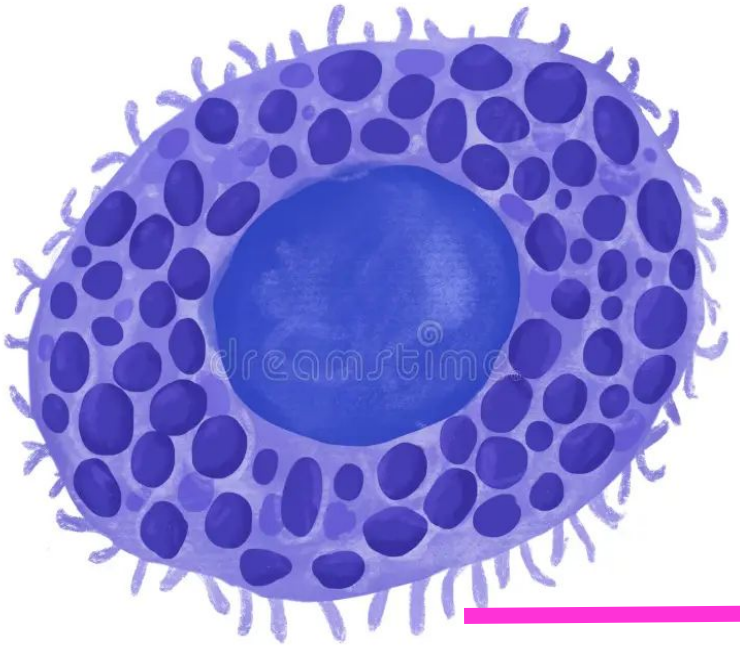

Immature CD8+ T cell
Antigen
Antigen Presenting Cell
TCR
MHCI
CD8+
Mature cytotoxic T Cell (Tc)


Classically activated macrophage (M1)
MICROBES
INF-γ
ROS, NO, lysosomal enzymes
IL-1, IL-12, IL-23, chemokines
Microbicidal actions: phagocytosis and killing of many bacteria and fungi
Inflammation
BOTH PRO-INFLAMMATORY AND ANTI-INFLAMMATORY
Alternatively activated macrophage (M2)
IL-4
IL-13
Growth factors, TGF-β
IL-10, TGF-β
Tissue repair, fibrosis
Anti-inflammatory effects

Lymphocytes



Mast Cells



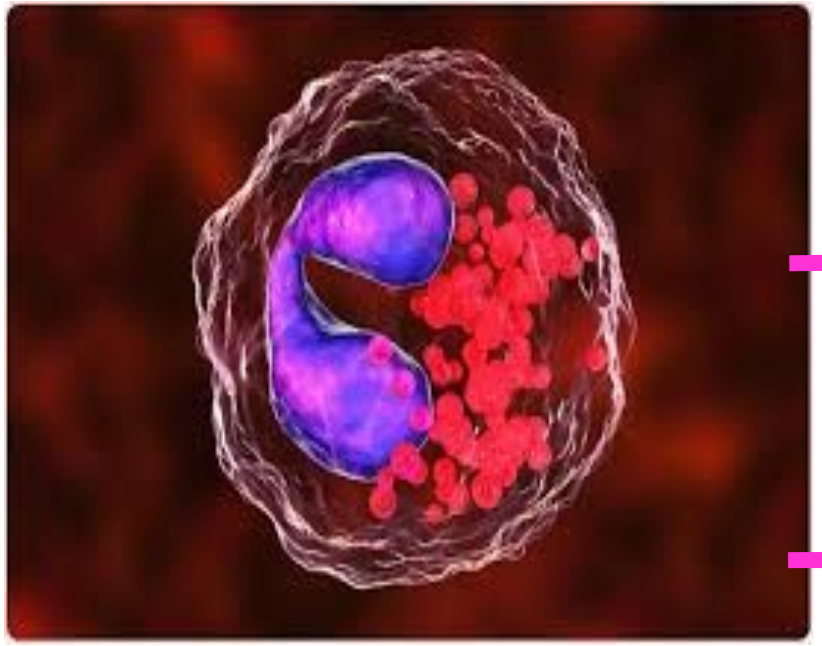
Histamine, Serotonin, Platelet activating factor → Increased vascular permeability

Prostaglandins → Vasodilatation, fever

Leukotrienes → Chemotaxis

Cytokines → Tissue repair

Eosinophils

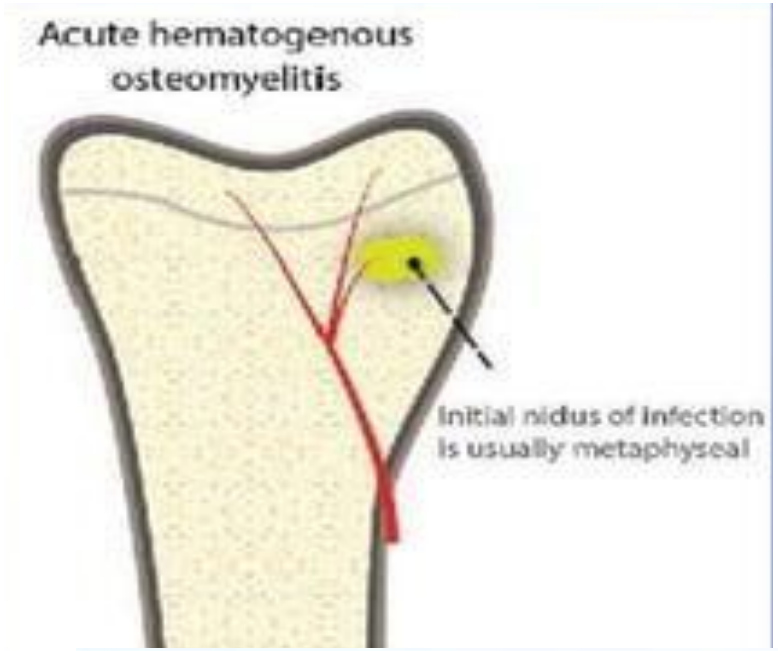


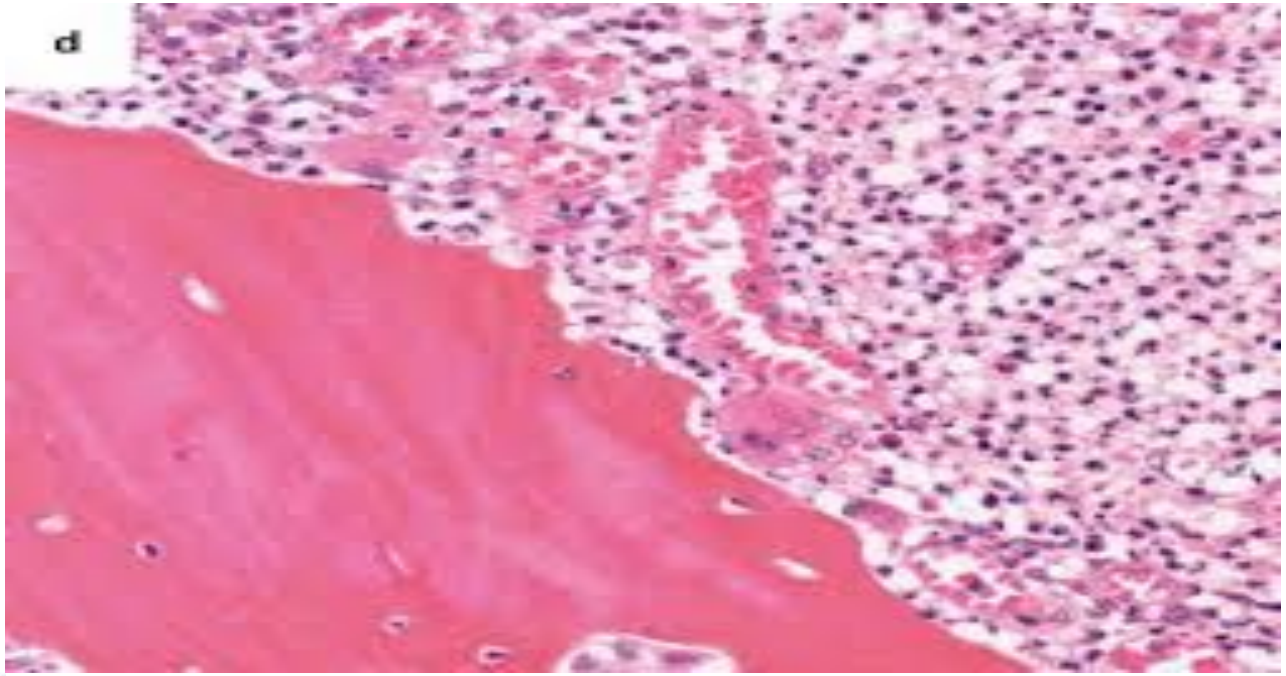
Cytokines → Activation of Mast cells, T-cells
B-cells
Tissue repair
Immune response regulation

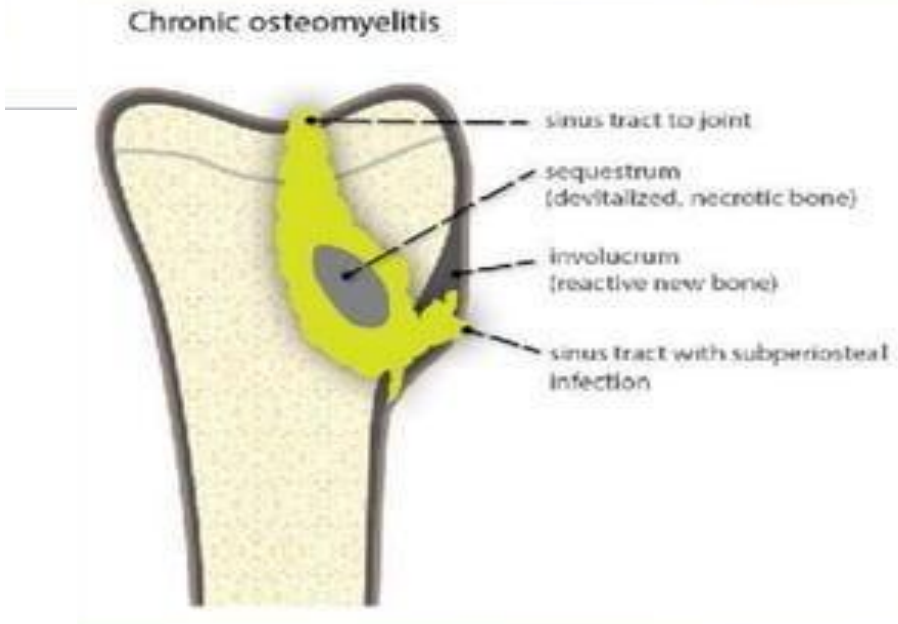
→ Phagocytosis

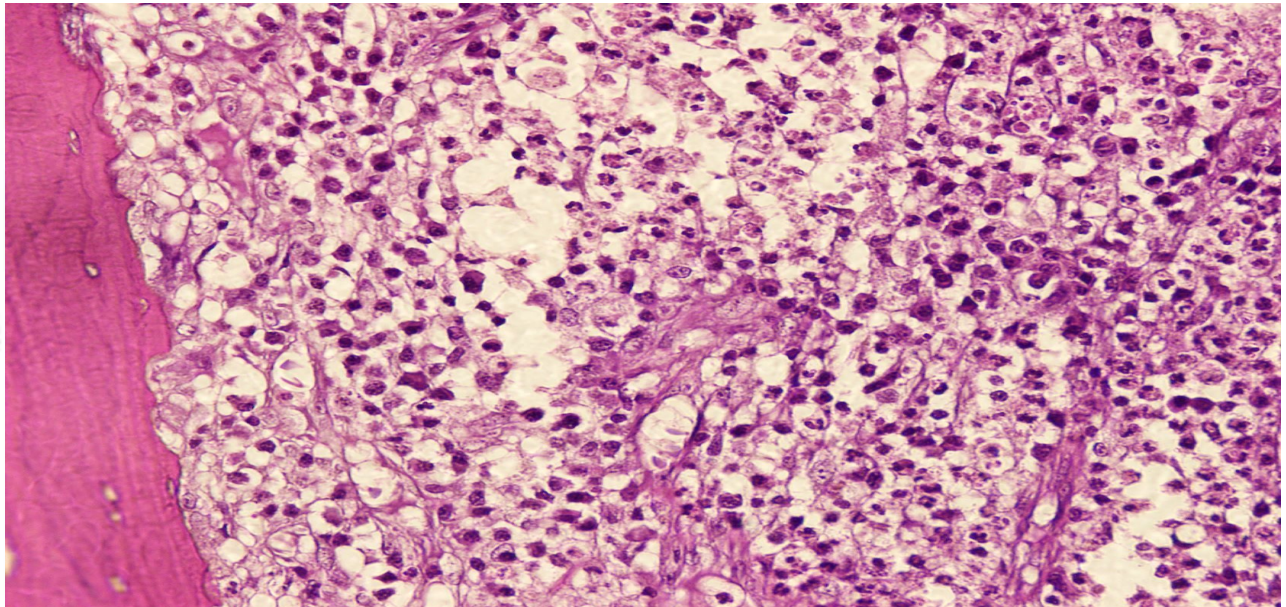
Cases of Chronic Inflammation

Following Acute Inflammation

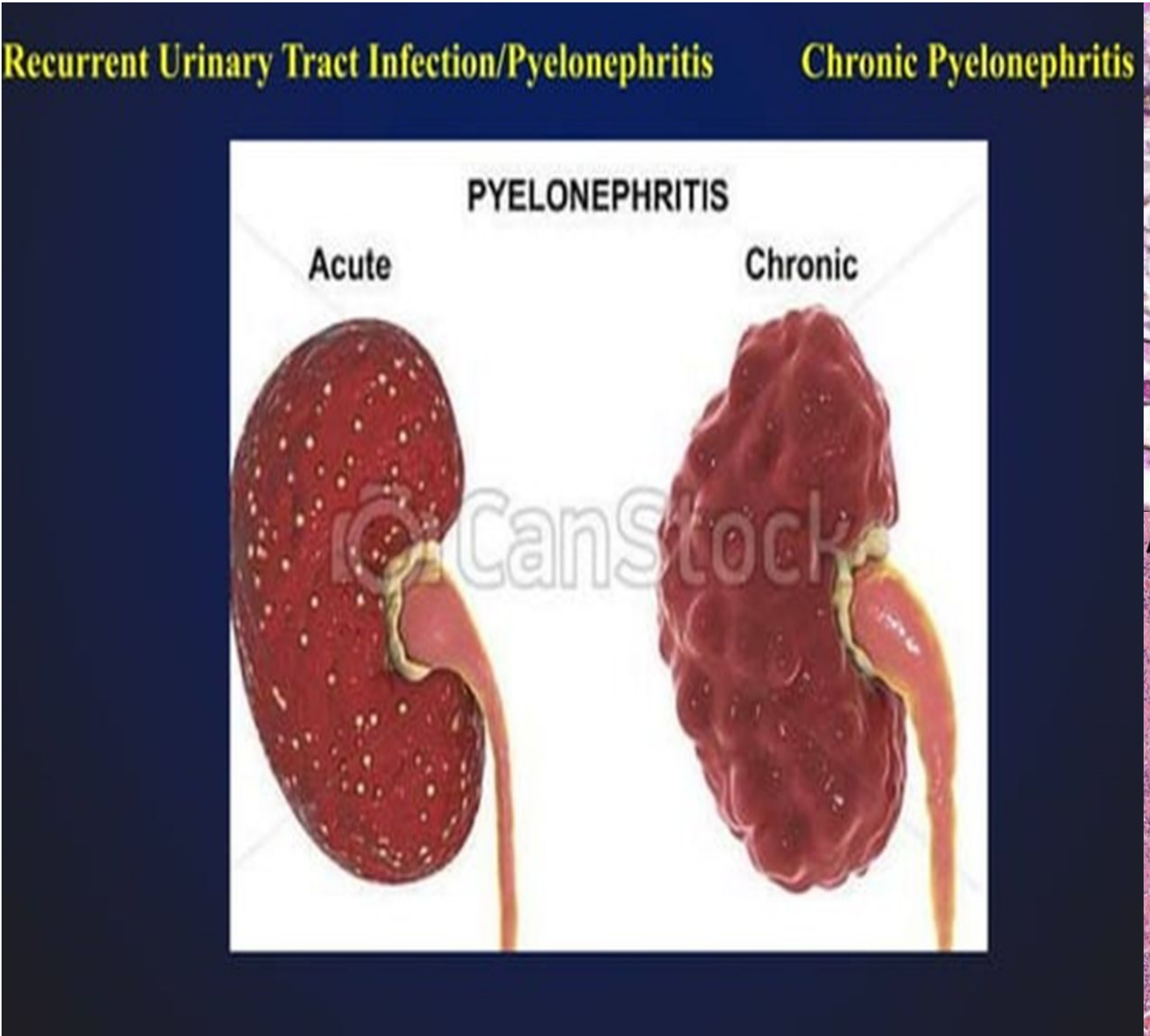


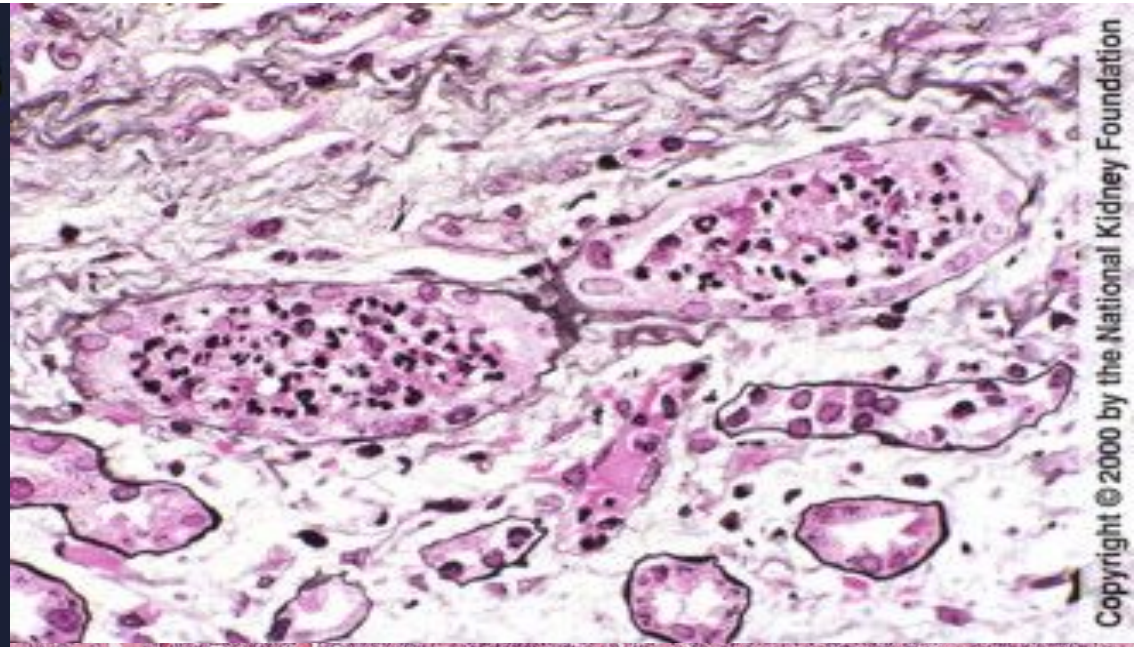


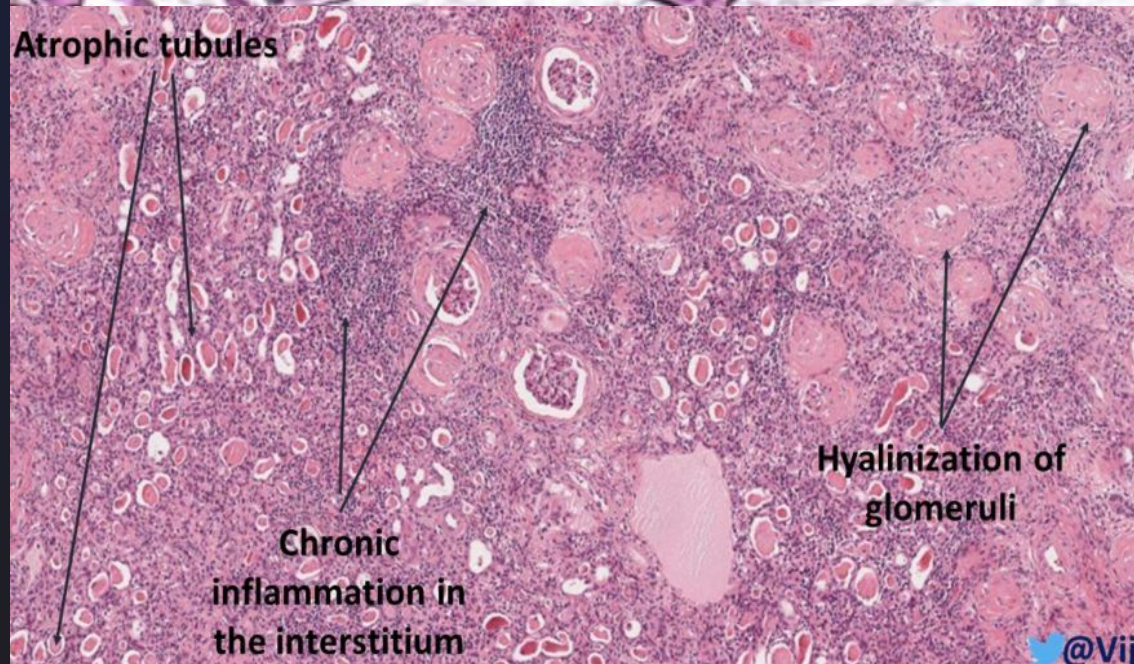




Recurrent Acute Inflammation

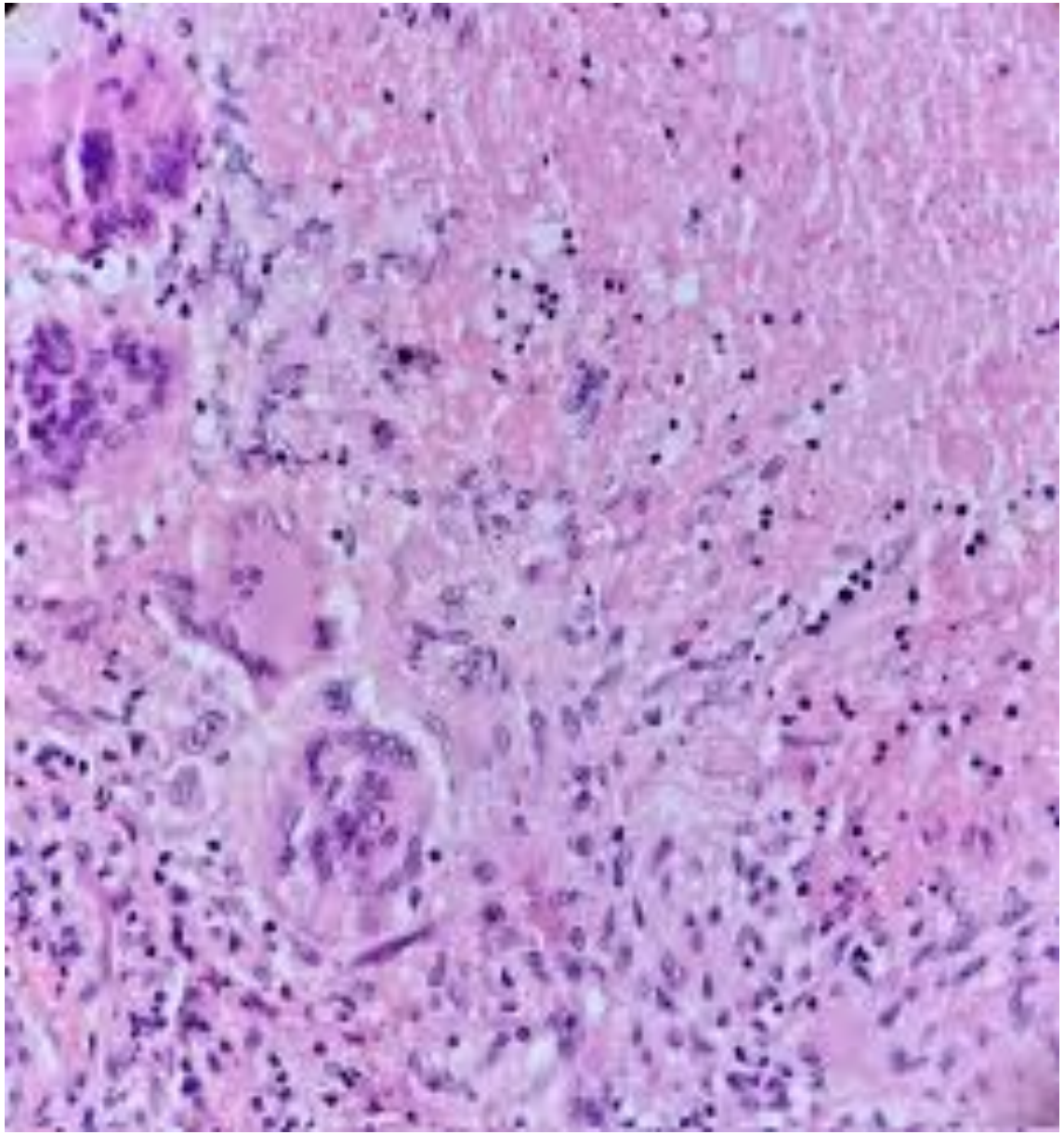






Chronic Inflammation starting de novo





Types of Chronic Inflammation

Chronic non-specific Inflammation

Granulomatous Inflammation

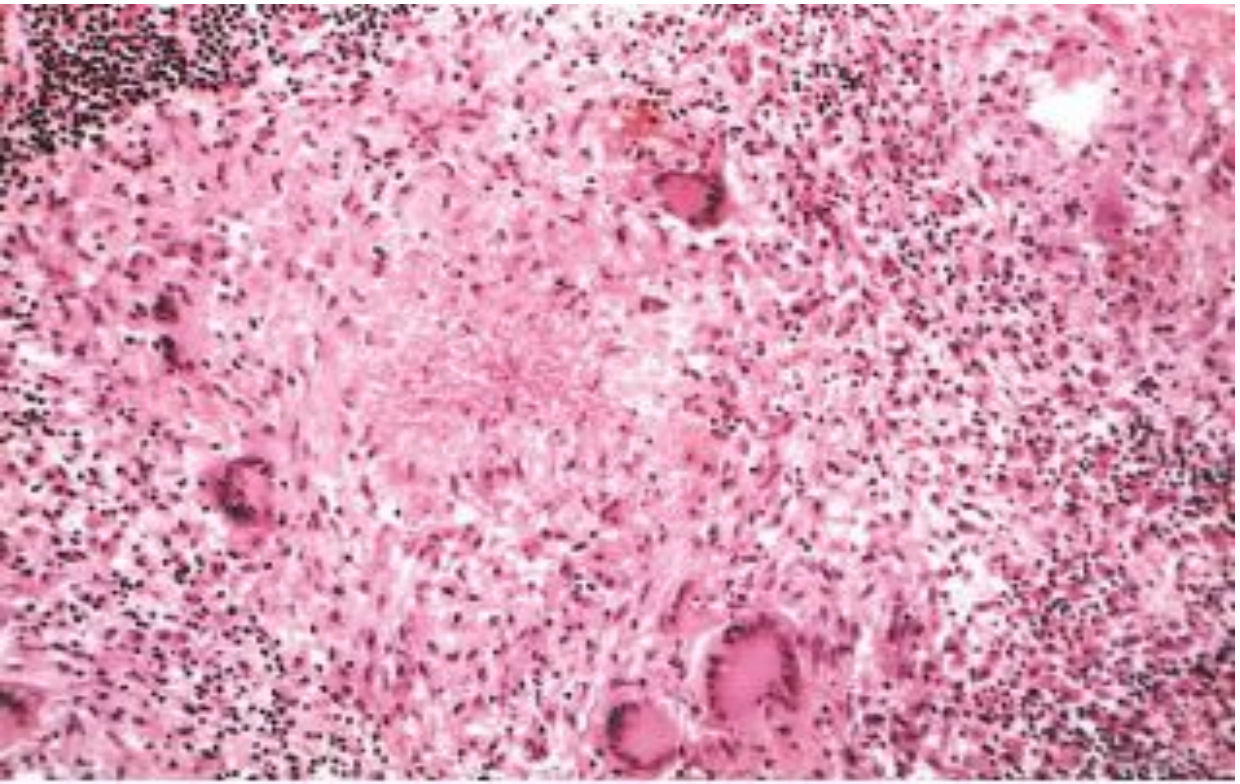
Chronic Non-Specific Inflammation

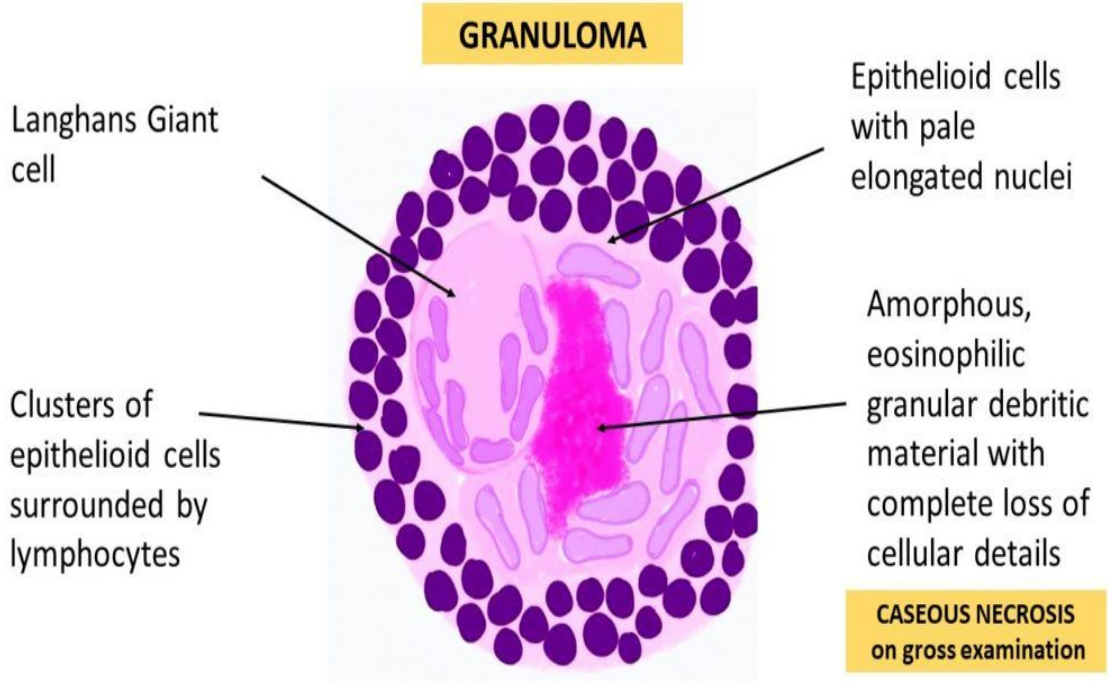
It is a type of chronic inflammation (prolonged response to a stimulus in which inflammation, tissue damage and attempt to repair coexist in a varying combinations) in which there is no granuloma formed. They are not histologically distinctive unlike specific chronic inflammation in which granuloma formation makes them histologically distinctive

Chronic Granulomatous Inflammation

Granulomatous inflammation is a form of chronic inflammation characterized by collections of activated macrophages, often with

T- lymphocytes, and sometimes associated with central necrosis





Chronic Granulomatous Inflammation

Disease	Cause	Tissue Reaction
Tuberculosis	*Mycobacterium tuberculosis*	Caseating granuloma (tubercle): focus of activated macrophages (epithelioid cells), rimmed by fibroblasts, lymphocytes, histiocytes, occasional Langhans giant cells; central necrosis with amorphous granular debris; acid-fast bacilli
Leprosy	*Mycobacterium leprae*	Acid-fast bacilli in macrophages; noncaseating granulomas
Syphilis	*Treponema pallidum*	Gumma: microscopic to grossly visible lesion, enclosing wall of histiocytes; plasma cell infiltrate; central cells are necrotic without loss of cellular outline
Cat-scratch disease	Gram-negative bacillus	Rounded or stellate granuloma containing central granular debris and recognizable neutrophils; giant cells uncommon
Sarcoidosis	Unknown etiology	Noncaseating granulomas with abundant activated macrophages
Crohn disease (inflammatory bowel disease)	Immune reaction against intestinal bacteria, possibly self antigens	Occasional noncaseating granulomas in the wall of the intestine, with dense chronic inflammatory infiltrate

GIANT CELLS



Cells with nuclei are arranged in 'HORSESHOE' shaped pattern. Can be seen in one or both the poles

Langhans

NOT Langerhans!



Contain regular nuclei scattered throughout cytoplasm

Foreign body



Ring of nuclei surrounding central cytoplasm. peripheral cytoplasm is vacuolated

Touton



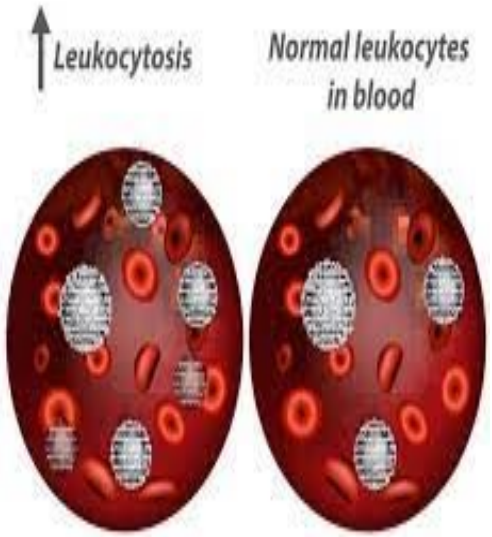
Contain hyperchromatic pleomorphic nuclei scattered throughout cytoplasm

Tumor

@VijayPatho

Systemic Effects of Inflammation





Acute phase reactant	Role
Dramatic increases in concentration:	
C-reactive protein Mannose binding protein	Fixes complement and opsonizes by coating with C3b
Moderate increases in concentration:	
α 1 anti-chymotrypsin C3, C9, factor B Ceruloplasmin Fibrinogen	Inhibit bacterial proteases increase complement function $+O_2^-$ scavenger Coagulation

Acute vs chronic inflammation

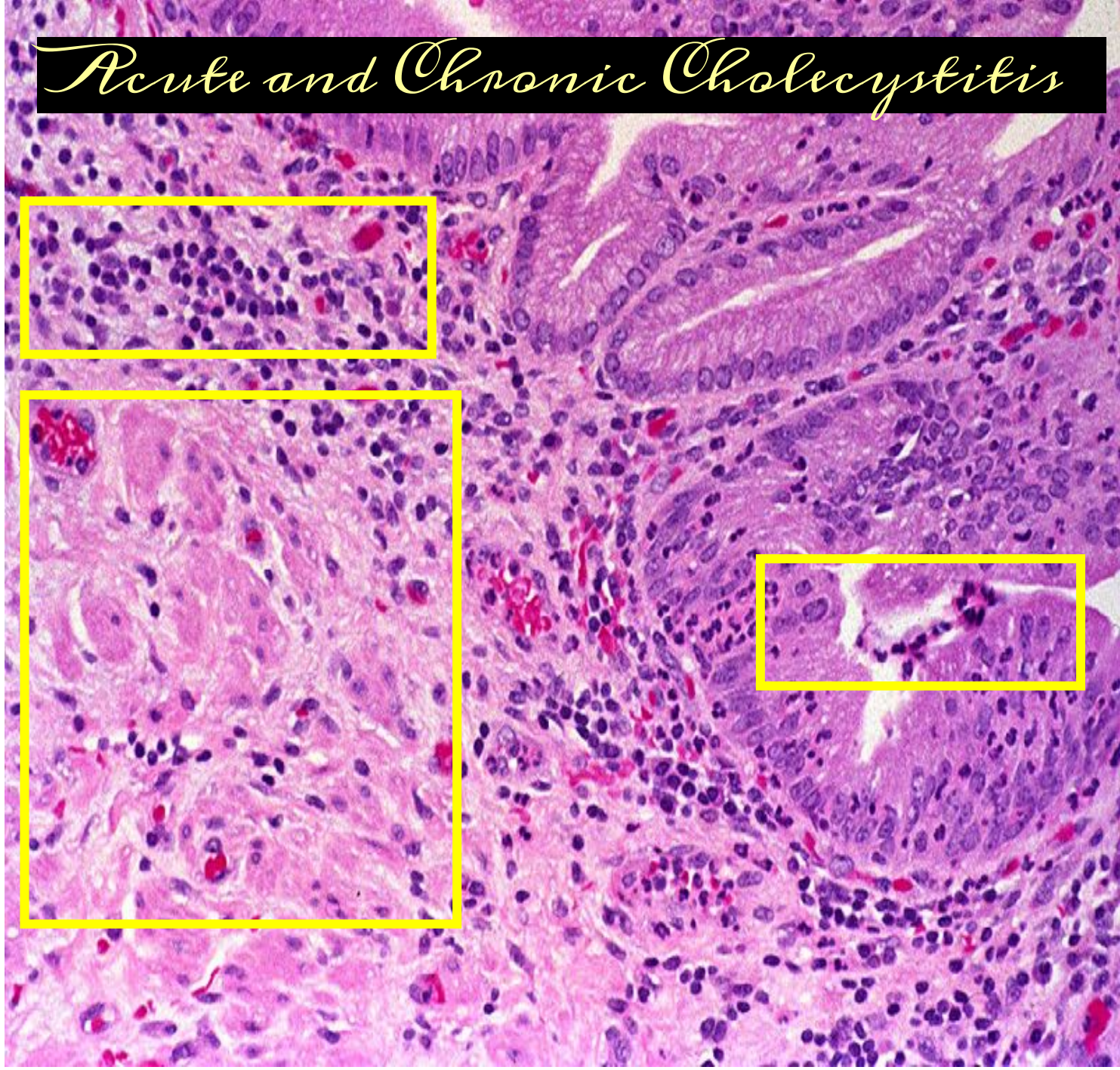
	ACUTE	CHRONIC
Causative agent	Pathogens, irritants, damage	Persistent acute inflammation due to non-degradable pathogens, persistent foreign bodies, or autoimmune reactions
Major cells	neutrophils, basophils, eosinophils, monocytes, macrophages	Monocytes, macrophages, lymphocytes, plasma cells, fibroblasts
Onset	Immediate	Delayed
Duration	Few days	Up to many months, or years
Outcomes	Resolution, abscess formation, chronic inflammation	Tissue destruction, fibrosis

73-year-old female presented with right upper quadrant pain and elevated WBC count. Physical exam revealed right upper quadrant tenderness. Ultrasound examination of the gallbladder revealed gallstones

The gallbladder was thickened and hyperemic. No localized masses were seen. The gallbladder contained numerous gallstones



The gallbladder mucosa is shows many neutrophils in the epithelium and lamina propria. The epithelium is reactive with enlarged nuclei and abundant eosinophilic cytoplasm and ulcerated in some areas. There is hemorrhage and fibroblastic proliferation throughout the wall. Plasma cells and macrophages are seen in the mucosa. No tumor is seen



Which of the following is the hallmark cell of chronic inflammation?

a) Neutrophil
b) Eosinophil
c) Macrophage
d) Basophil

Granulomatous inflammation is a subtype of chronic inflammation characterized by:

a) Neutrophilic infiltration
b) Aggregates of activated macrophages
c) Widespread necrosis
d) Edema formation

Which of the following is NOT a cause of chronic inflammation?

a) Persistent infections
b) Prolonged exposure to toxic agents
c) Acute trauma
d) Autoimmune diseases

Which cytokine is most closely associated with macrophage activation in chronic inflammation?

a) IL-10
b) IL-14
c) IFN-γ
d) TNF-α

Langhans-type giant cells are most commonly seen in:

a) Acute suppurative inflammation
b) Granulomatous inflammation
c) Serous inflammation
d) Allergic inflammation

Which of the following is a classic example of chronic granulomatous inflammation?

a) Acute appendicitis
b) Tuberculosis
c) Pneumonia
d) Angina pectoris

The resolution of chronic inflammation involves all EXCEPT:

a) Removal of offending agent
b) Regeneration or scar formation
c) Mast cell degranulation
d) Downregulation of pro-inflammatory cytokines

A 55-year-old male patient visits the oral medicine clinic with a non-healing ulcer on the lateral border of his tongue, persisting for more than 6 weeks. He is a chronic smoker and consumes alcohol occasionally. The lesion is indurated, with raised margins and surrounding erythema. There is mild pain, especially while eating spicy food. On palpation, there is a firm submandibular lymph node. An incisional biopsy is taken, and histopathology reveals chronic inflammatory infiltrate with lymphocytes and plasma cells, along with epithelial dysplasia.

1-What is the clinical term for a non-healing sore in the oral cavity?

-Ulcer

2-What type of inflammation is seen in this lesion?

-Chronic

3-What is the main type of immune cell found in chronic inflammation?

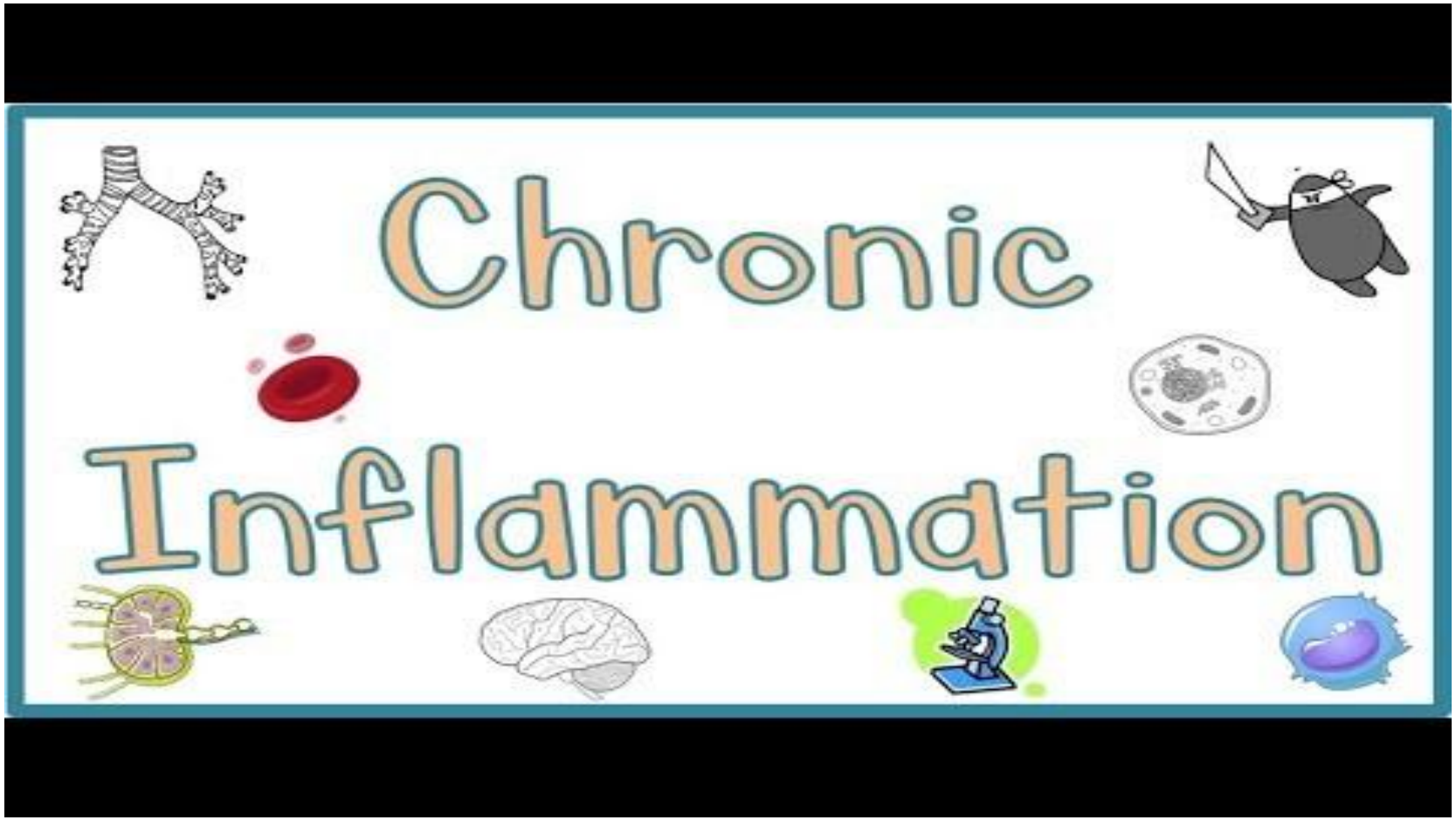
-Macrophage

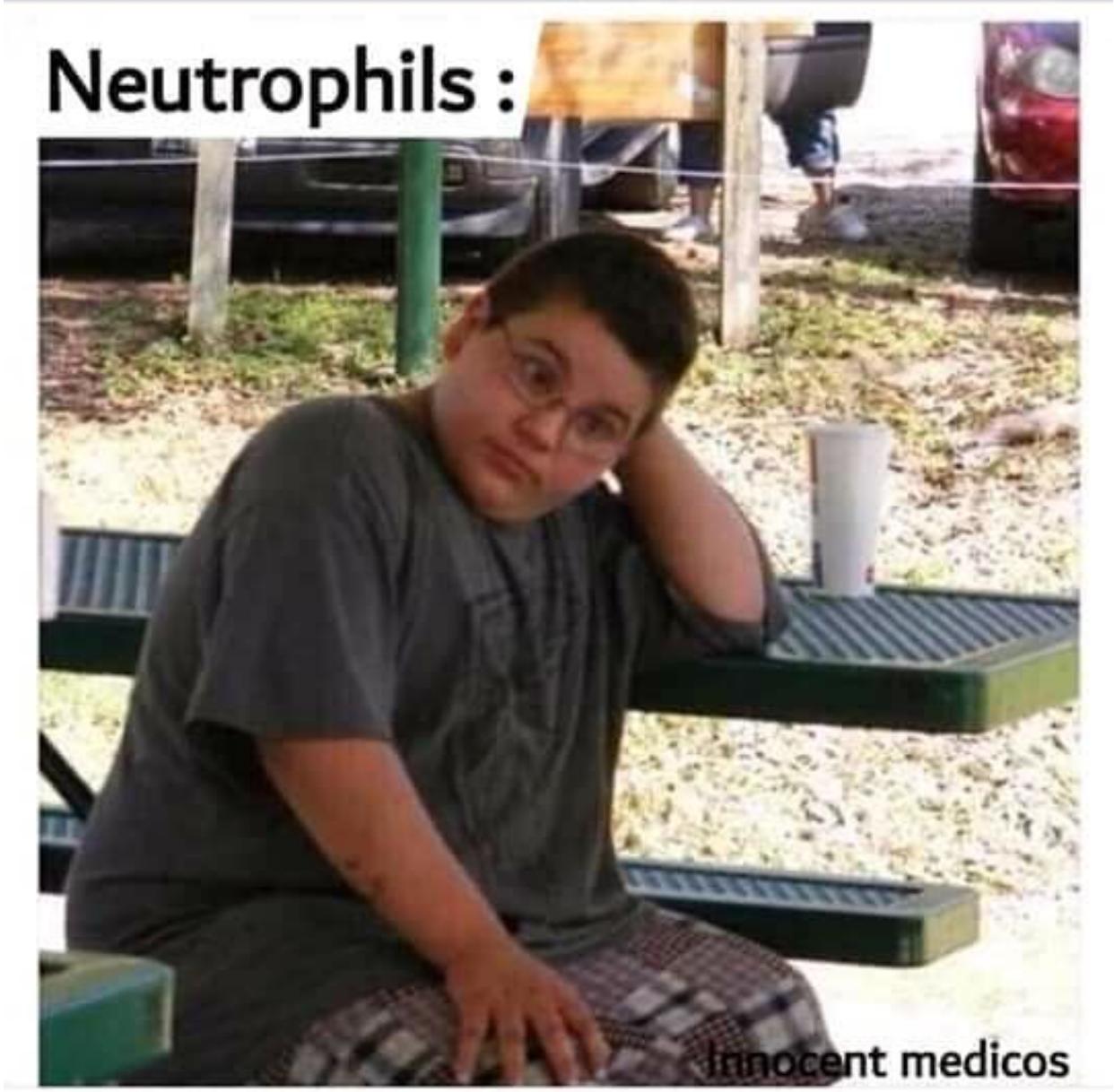
4-Which epithelial change was noted in the biopsy?

-Dysplasia

Additional Read



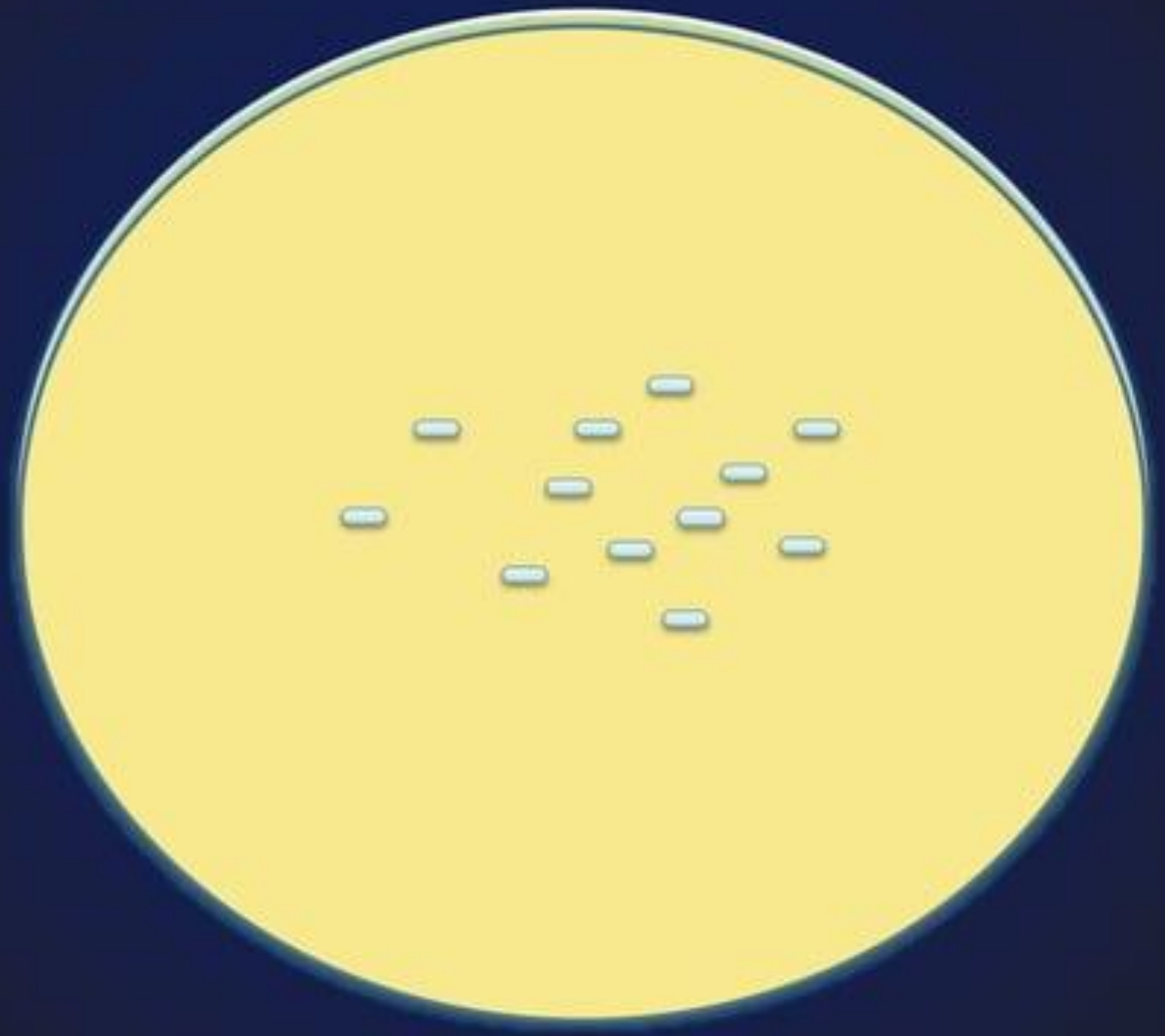


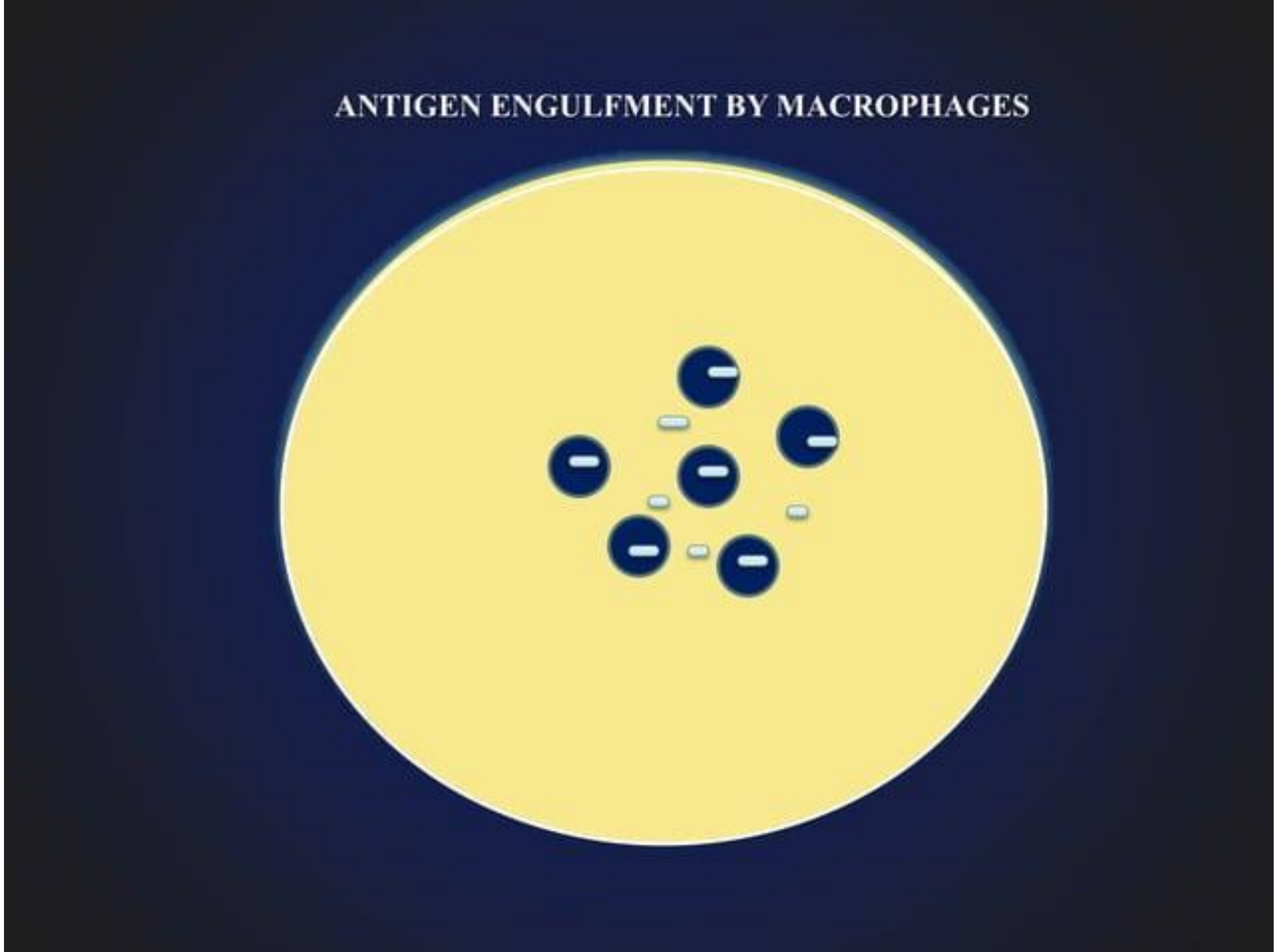


Infection control nai horha hamsey. Help leni paareygi.

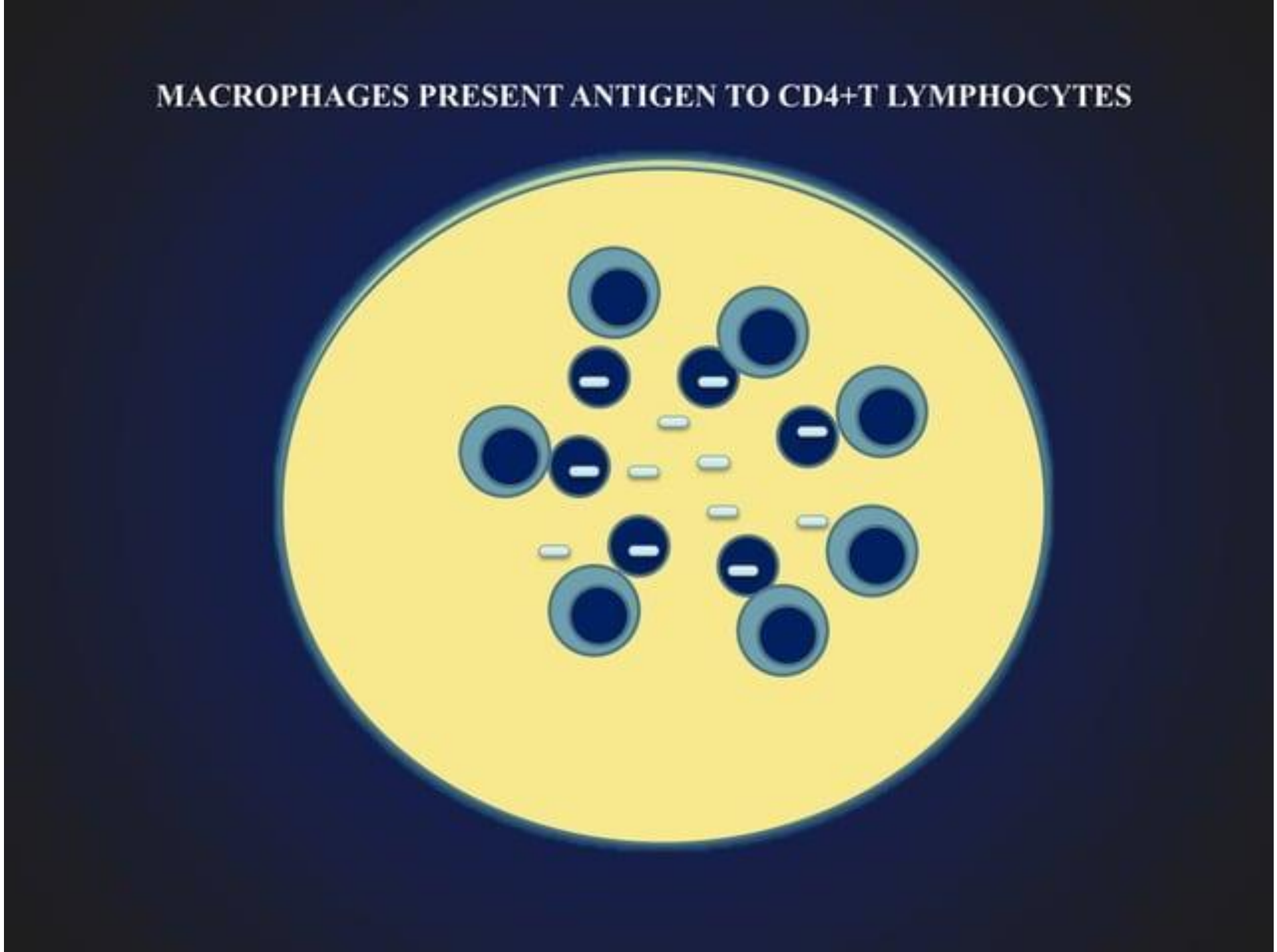
Thank You

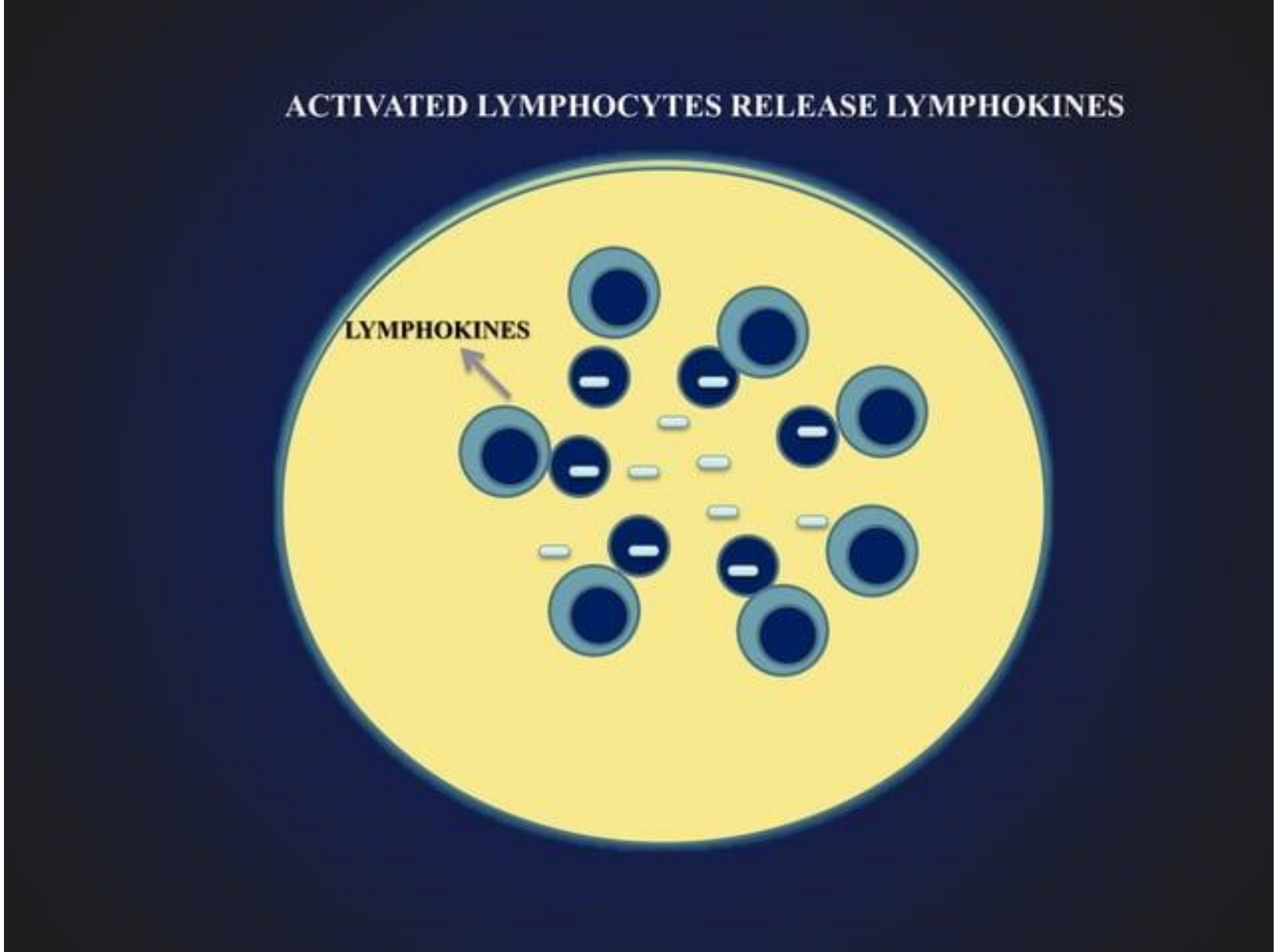
ANTIGEN ENTRY INTO TISSUE

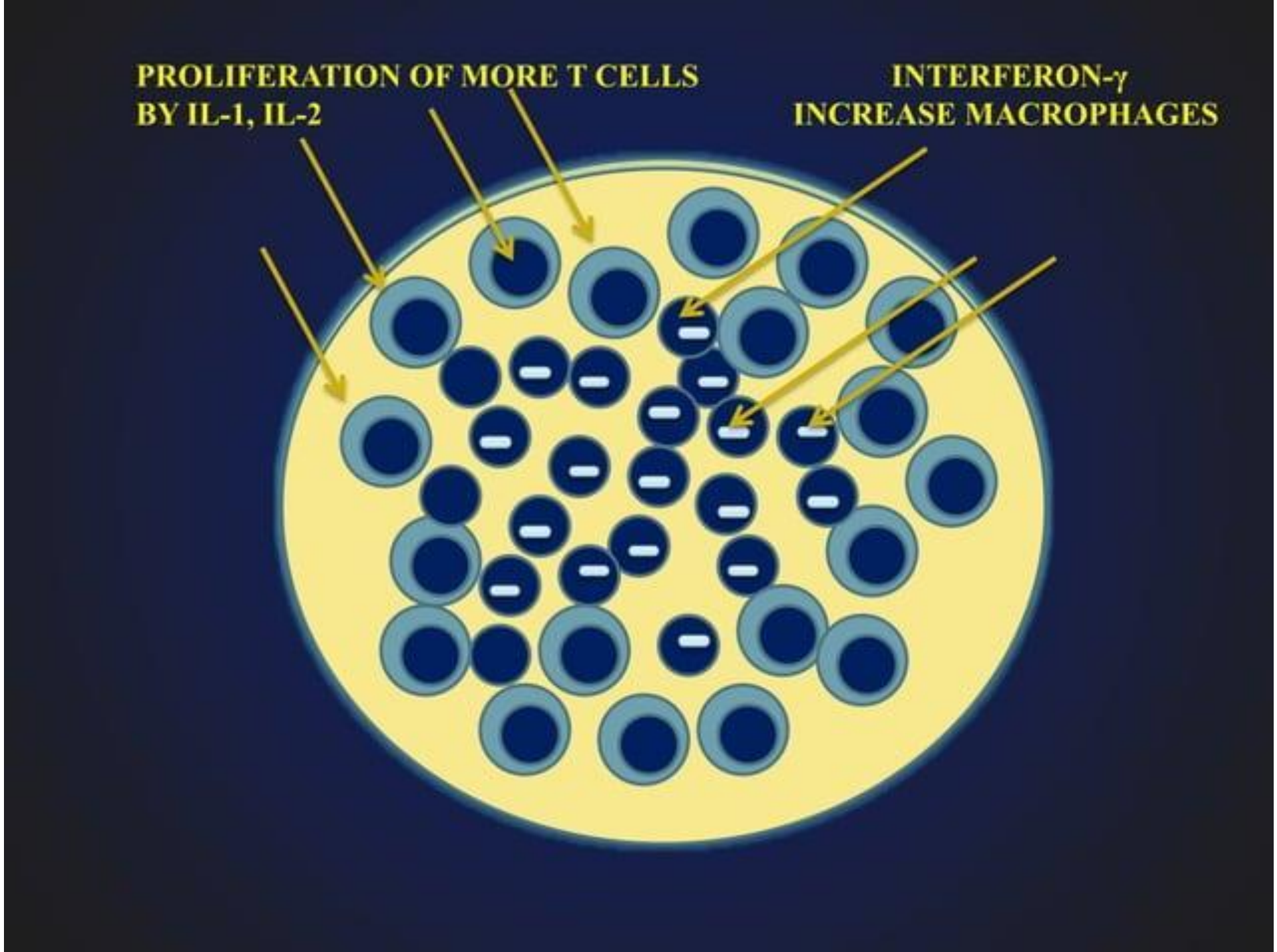



ANTIGEN ENGULFMENT BY MACROPHAGES

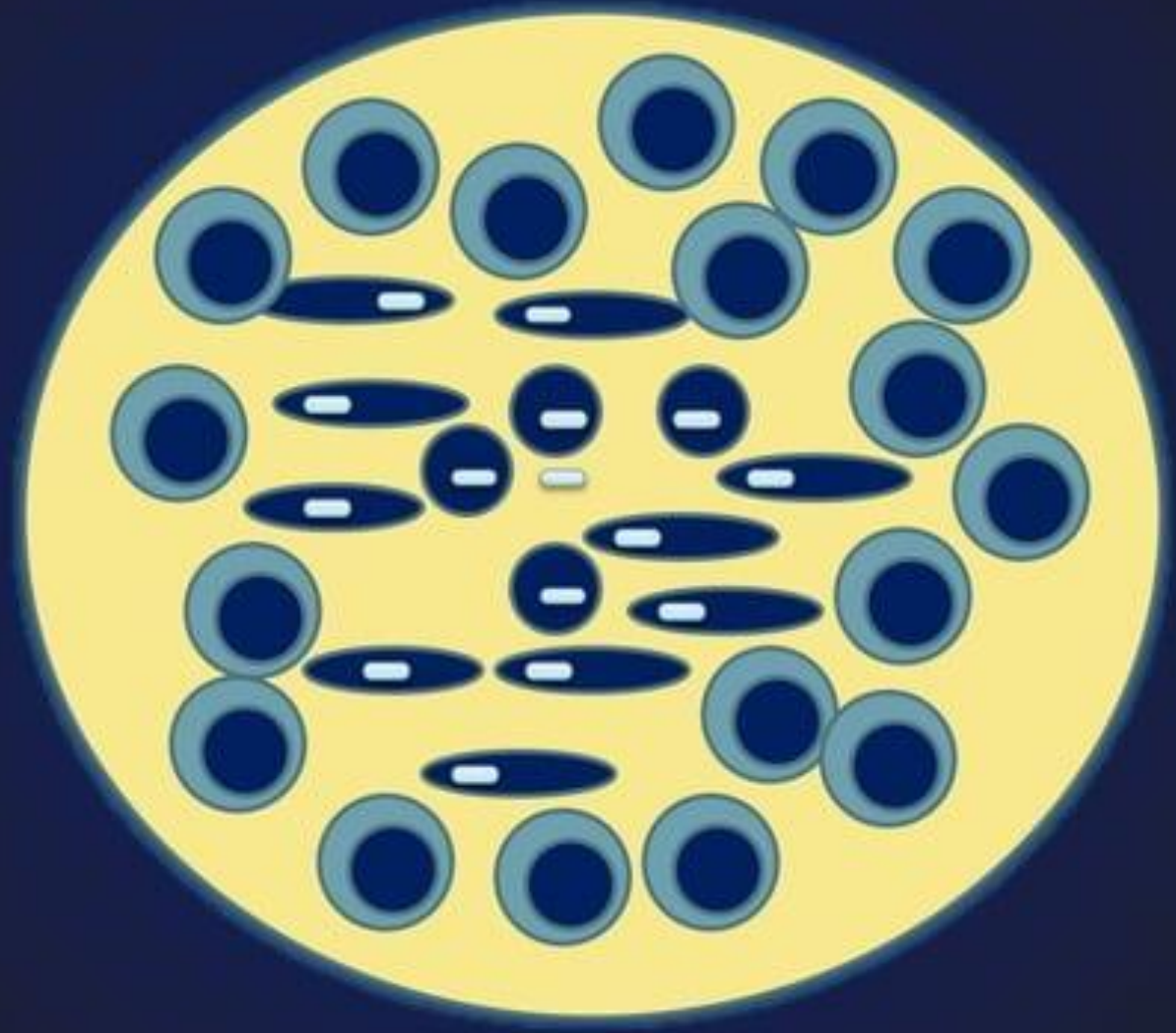
MACROPHAGES PRESENT ANTIGEN TO CD4+T LYMPHOCYTES



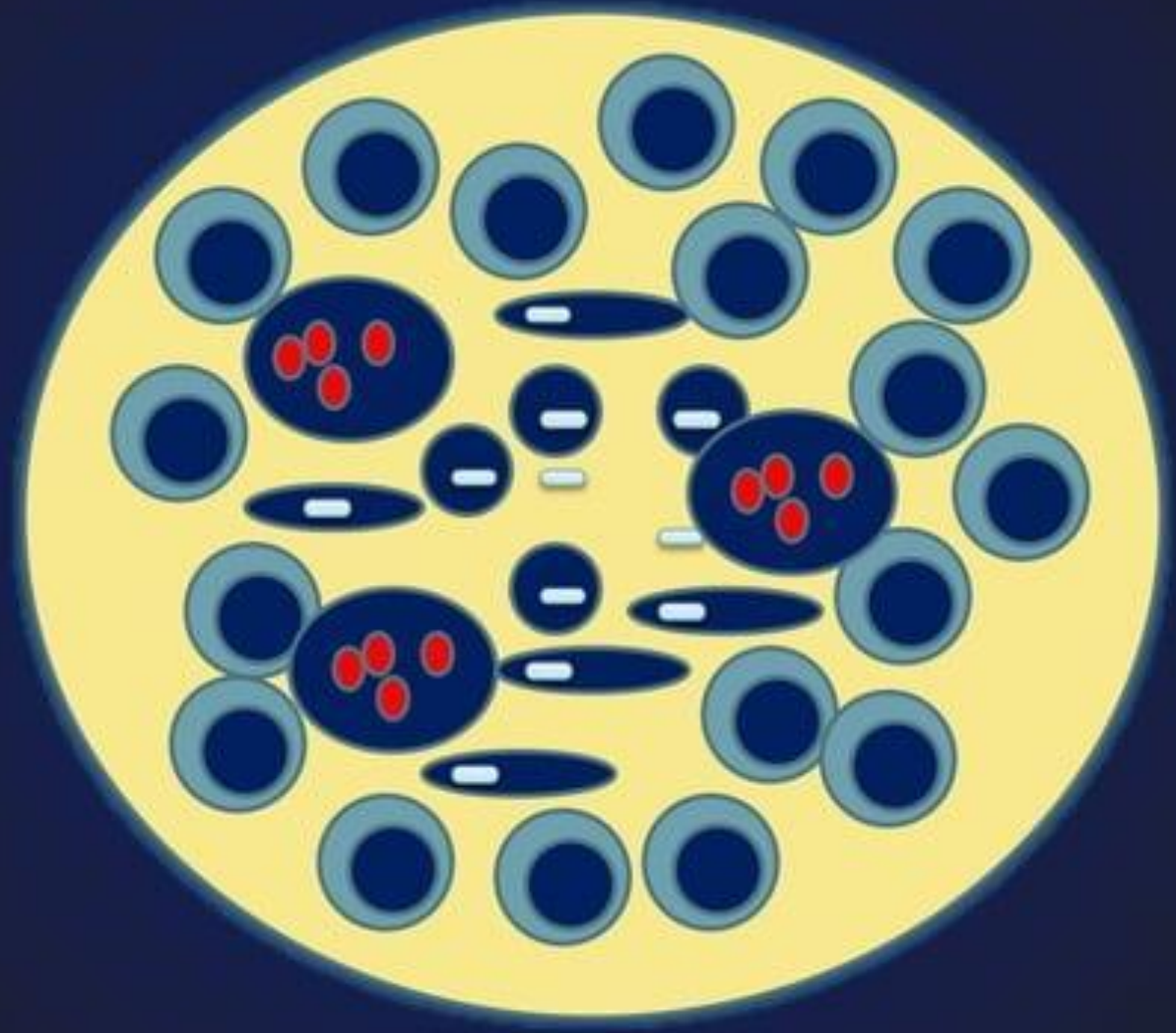

ACTIVATED LYMPHOCYTES RELEASE LYMPHOKINES
LYMPHOKINES


PROLIFERATION OF MORE T CELLS
BY IL-1, IL-2
INTERFERON-γ
INCREASE MACROPHAGES

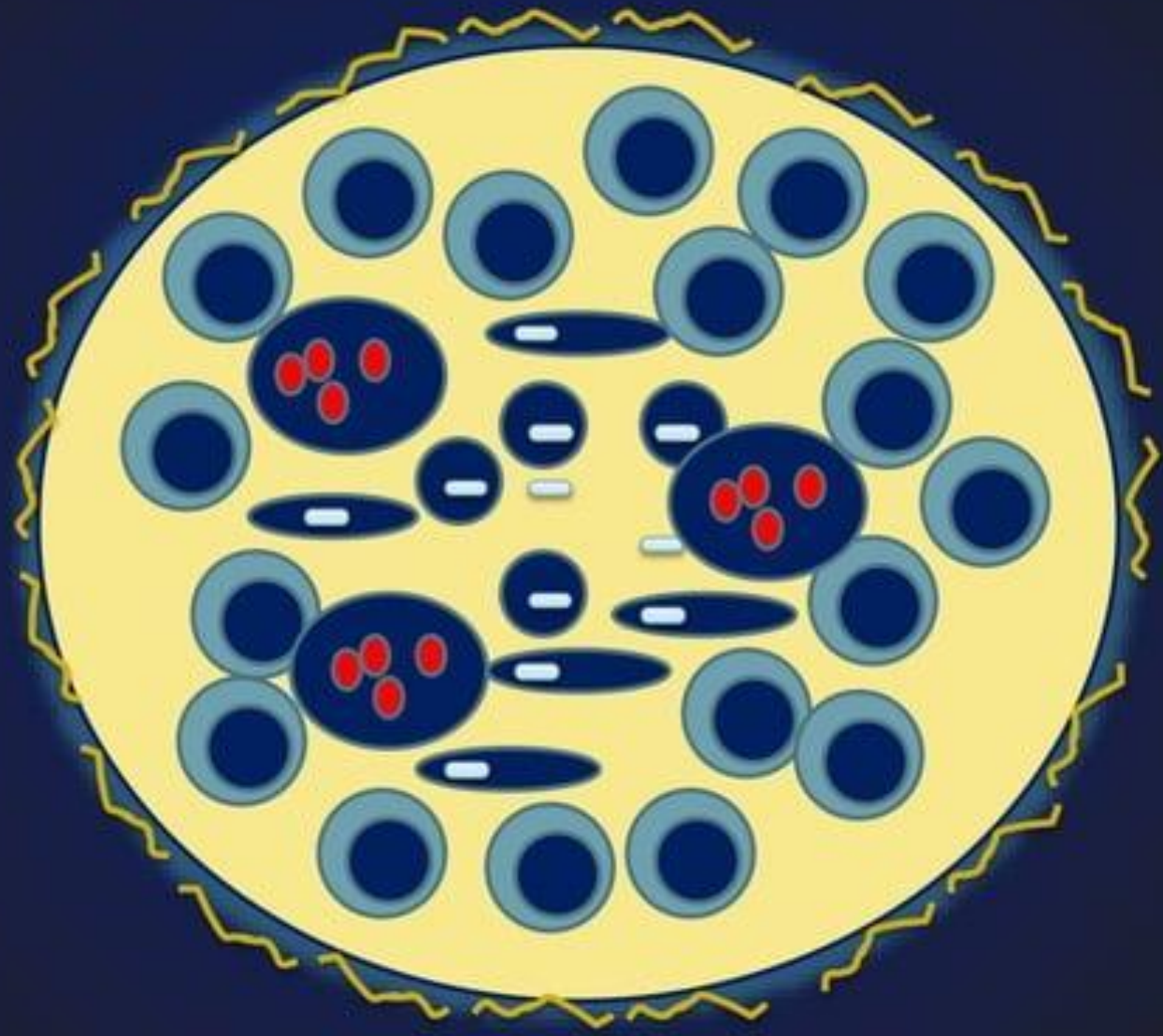
ACTIVATED MACROPHAGES TRANSFORM TO
EPITHELIOID CELLS

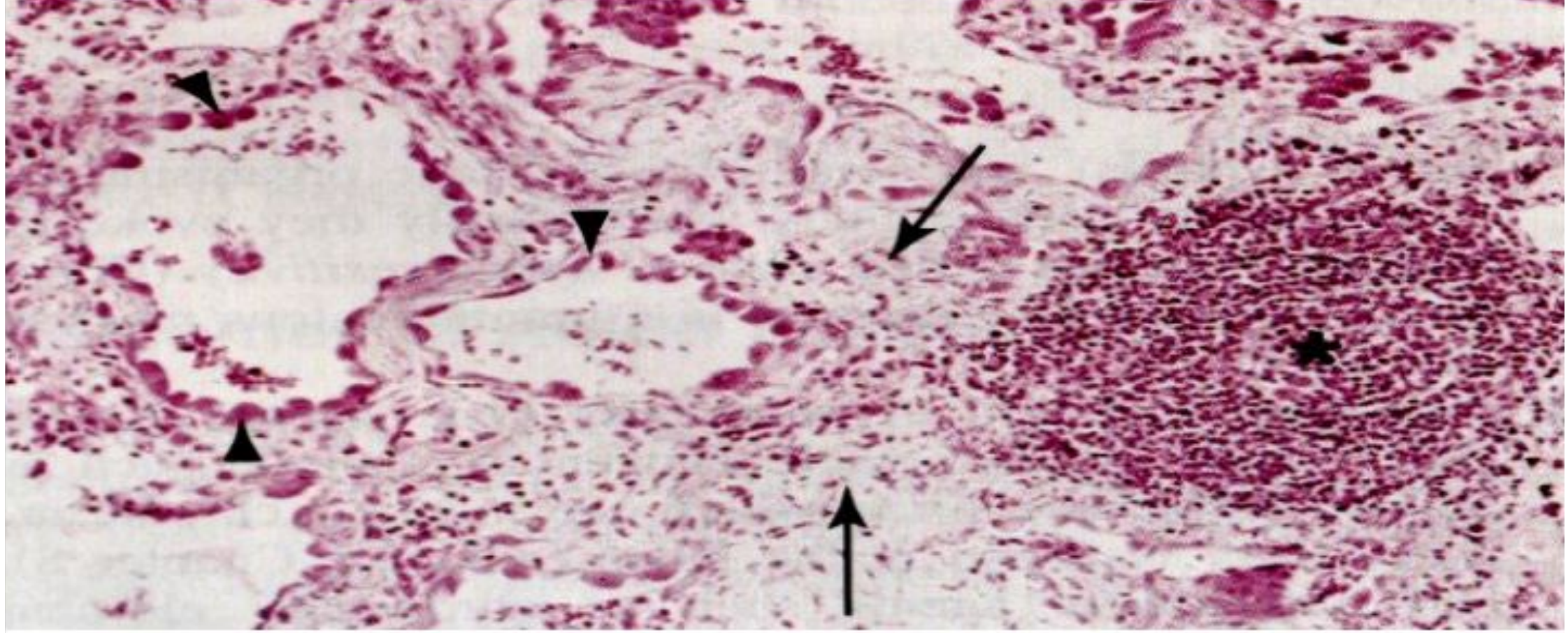


EPITHELIOID CELLS FUSE TO FORM
MULTINUCLEATED GAINT CELLS



HEALING BY FIRBOBLASTS





The three characteristic features of chronic inflammation (in the lung)

1. **Chronic inflammatory cells infiltration.***
2. **Destruction of the normal tissue.**

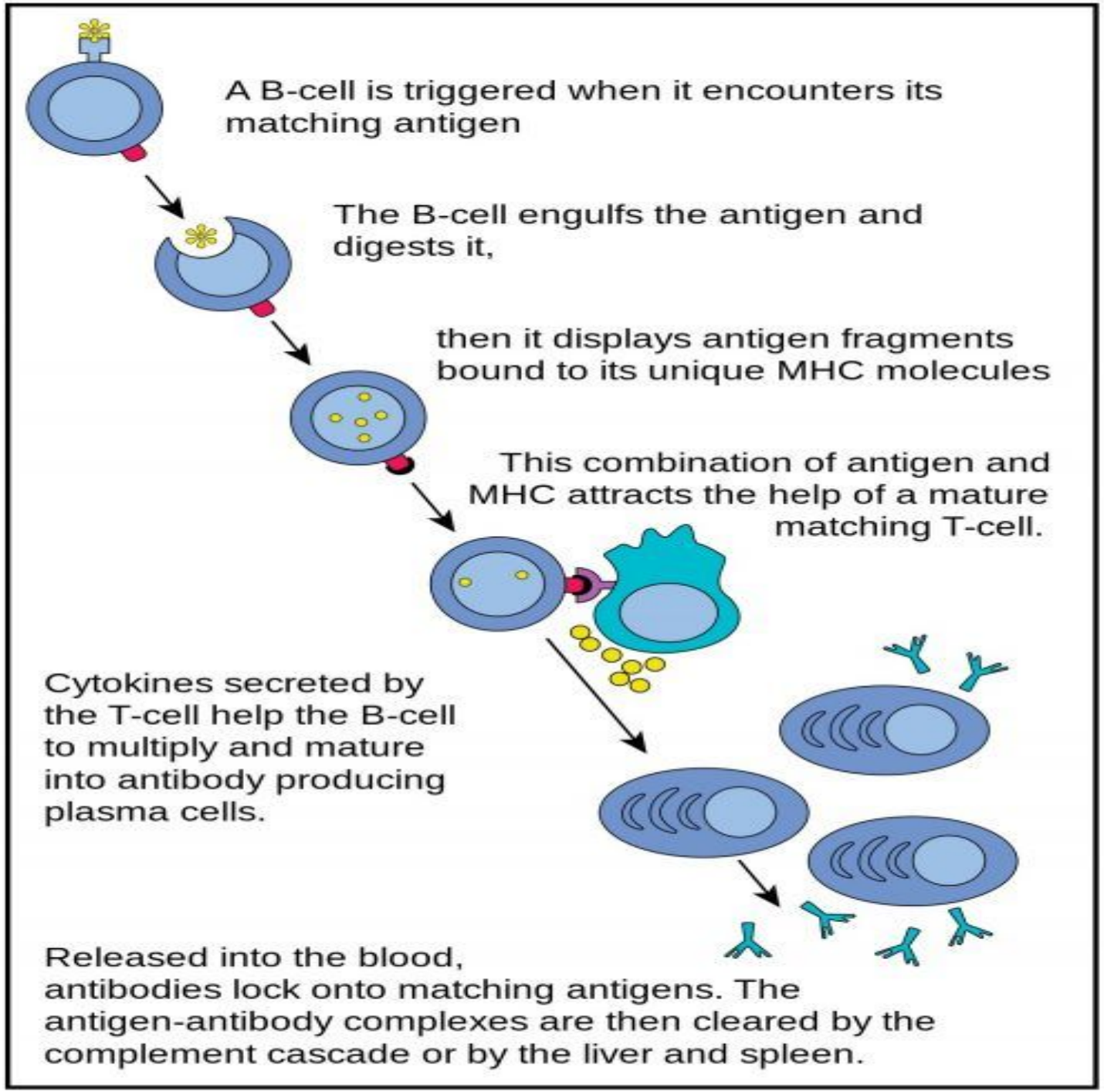
(normal alveoli are replaced by spaces lined by cuboidal cells (arrow heads).

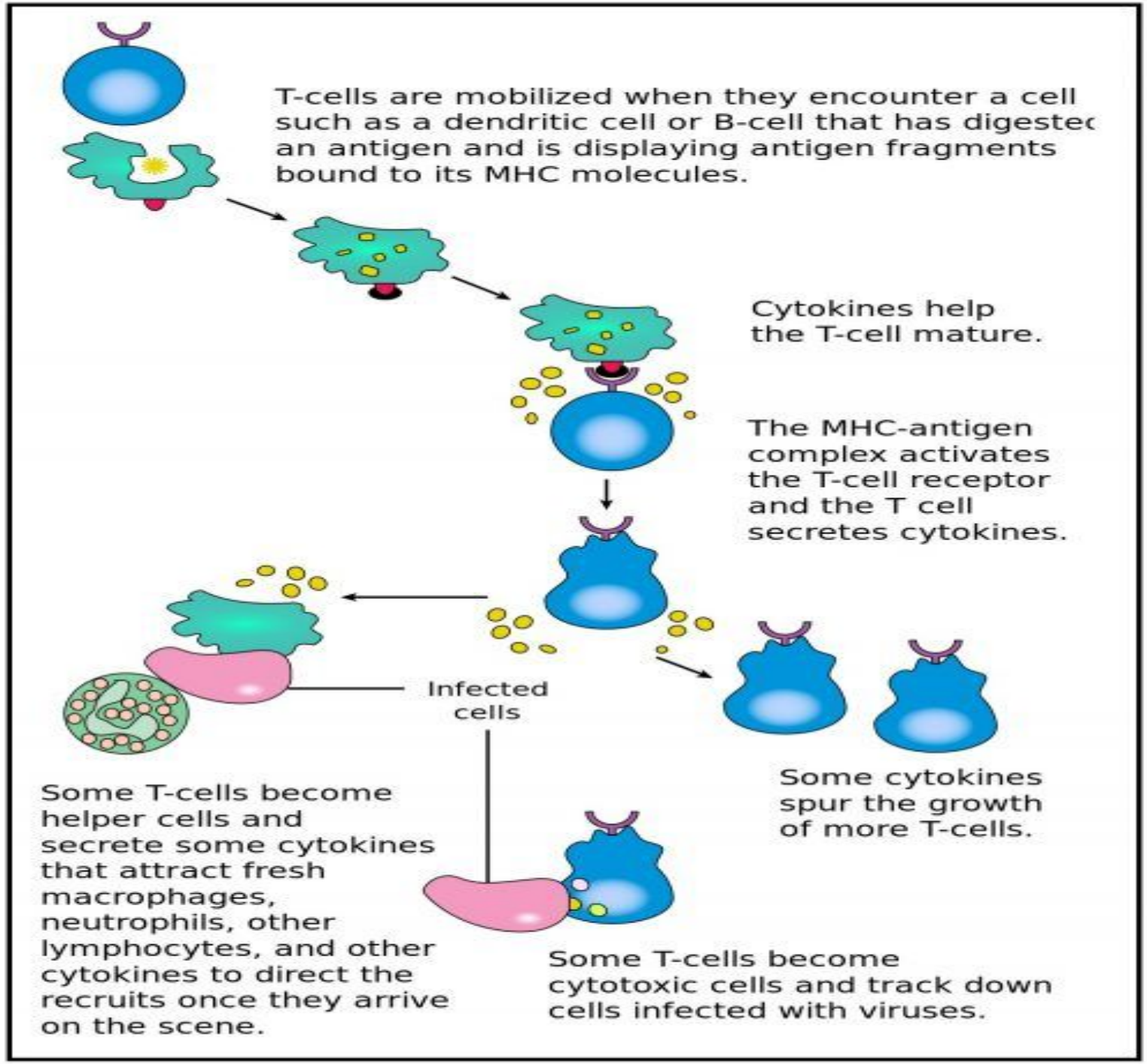
3. **Replacement by fibrosis (arrows).**

Mediators of chronic inflammation

Agent	Action	Source
Migration inhibition factor (MIF)	Aggregation of macrophages at site of injury	Activated T lymphocytes
Macrophage activating factor (MAF)	Increased phagocytosis by macrophages	Activated T lymphocytes
Complement C5a	Chemotatic for macrophages	Complement system
Eosinophil chemotactic factor of anaphylaxis (ECF A)	Chemotactic for eosinophils in metazone infection	Mast cells and basophils

Some
symptoms
of chronic
inflammation


A B-cell is triggered when it encounters its matching antigen
The B-cell engulfs the antigen and digests it,
then it displays antigen fragments bound to its unique MHC molecules
This combination of antigen and MHC attracts the help of a mature matching T-cell.
Cytokines secreted by the T-cell help the B-cell to multiply and mature into antibody producing plasma cells.
Released into the blood, antibodies lock onto matching antigens. The antigen-antibody complexes are then cleared by the complement cascade or by the liver and spleen.


T-cells are mobilized when they encounter a cell such as a dendritic cell or B-cell that has digested an antigen and is displaying antigen fragments bound to its MHC molecules.
Cytokines help the T-cell mature.
The MHC-antigen complex activates the T-cell receptor and the T cell secretes cytokines.
Infected cells
Some cytokines spur the growth of more T-cells.
Some T-cells become helper cells and secrete some cytokines that attract fresh macrophages, neutrophils, other lymphocytes, and other cytokines to direct the recruits once they arrive on the scene.
Some T-cells become cytotoxic cells and track down cells infected with viruses.